NOVA @examlix

PHYSICS

1) Two identical rings, each of radius R metres, are coaxially placed R metres apart. If Q_1 coulombs and Q_2 coulombs are the charges uniformly spread on the two rings respectively, then the work done in moving a charge q from the centre of one ring to that of the other is :–

(1) zero

(2) $\frac{q(Q_1-Q_2)(\sqrt{2}-1)}{\sqrt{2}(4\pi\varepsilon_0 R)}$

(3) $\frac{q\sqrt{2}(Q_1+Q_2)}{(4\pi\varepsilon_0 R)}$

(4) $\frac{q(Q_1+Q_2)(\sqrt{2}+1)}{\sqrt{2}(4\pi\varepsilon_0 R)}$

2) The surface charge density of a thin charged disc of radius R is σ. With respect to the field at the centre, the electric field along the axis at a distance $\sqrt{3}$R from the centre of the disc :

(1) reduces by 70.7%

(2) reduces by 29.3%

(3) reduces by 86.6%

(4) reduces by 13.4%

3) Three infinite line charges of charge per unit length λ, 2λ and –λ are kept in xy-plane parallel to the y-axis. They pass through x = –a, x = a and x = 2a respectively. Find the electric field at different points on the x-axis :-

(1) At x = –2a, $E = -\frac{25k\lambda}{6a}\hat{i}$

(2) At x = 0, $E = -\frac{2k\lambda}{a}\hat{i}$

(3) At x = $\frac{3a}{2}$, $E = \frac{64k\lambda}{5a}\hat{i}$

(4) At x = 0, $E = -\frac{2k\lambda}{3a}\hat{i}$

4) A cube is made from six thin insulating square faces, each square having side ‘d’ and carrying a uniformly distributed charge of Q. The magnitude of electric flux leaving the sixth face, due to the electric field of other five faces, is ϕ. The magnitude of electrostatic force on each face of cube is F. Then

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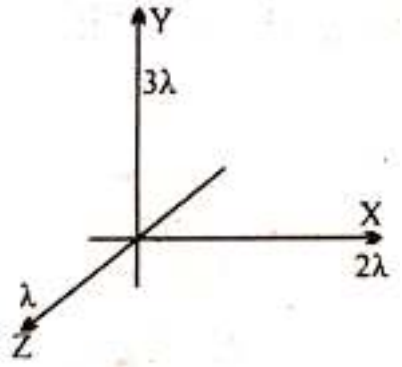
(1) $\phi = \frac{5Q}{6\epsilon_0}$

(2) $\phi = \frac{Q}{6\epsilon_0}$

(3) $F = \frac{Q^2}{2\epsilon_0 d^2}$

(4) $F = \frac{Q^2}{\epsilon_0 d^2}$

5) The diagram shows three infinitely long uniform line charges of charge per unit length $2\lambda, 3\lambda$ and λ Cm^{-1} placed on the X, Y and Z axes respectively. The work done in moving a unit positive charge from (1, 1, 1) m to (0, 1, 1) m is equal to (centers of the wires are coinciding with origin)



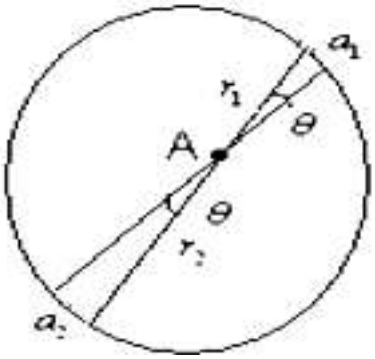
(1) $\frac{\lambda \ln 2}{2\pi\varepsilon_0}$

(2) $\frac{\lambda \ln 2}{\pi\varepsilon_0}$ J

(3) $\frac{3\lambda \ln 2}{2\pi\varepsilon_0}$ J

(4) Zero

6) A wire having uniform linear charge density λ, is bent in the form of a circle of radius R. Point A is in the plane of the ring but not at the center as shown in figure. Two elements of the ring of lengths a_1 and a_2 subtend very small and same angle at the point A. The elements are at distances r_1 and r_2 from the point A respectively and $r_2 > r_1$. Then choose the **INCORRECT** statement of the following



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The ratio of charges of elements a_1 and a_2 is

(1) $\frac{r_1}{r_2}$

(2) Magnitude of electric field produced by element a_1 at point A is greater than that produced by element a_2

(3) The elements a_1 and a_2 produce same potential at A

(4) Net electric field produced by the elements only at A is zero

7) Two equal point charges are fixed at x = –a and x = +a on the x-axis. Another point charge Q in placed at the origin. The change in electrical potential energy of Q when it is displaced by a small distance x along the x-axis is approximately proportional to

(1) x

(2) x^2

(3) x^3

(4) $\frac{1}{x}$

8) The electric field at 20 cm from the centre of a uniformly charged sphere of radius 10 cm is E. Then at a distance 5 cm from the centre it will be :-

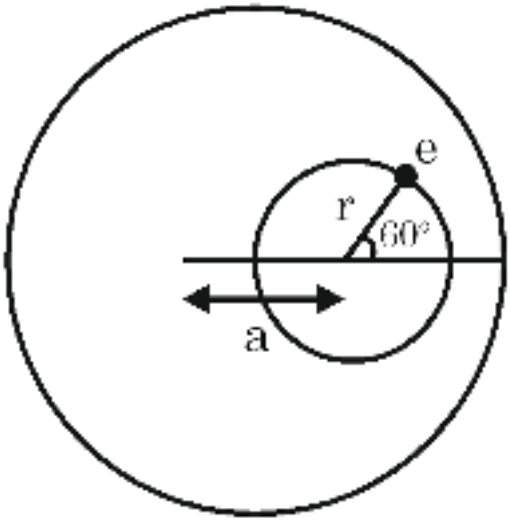
(1) 16 E

(2) 4 E

(3) 2 E

(4) zero

9) A cavity of radius r is present inside a solid dielectric sphere of radius R having volume charge density ρ, distance b/w centre of sphere and cavity is a. An electron 'e' is kept inside cavity at an angle $\theta = 60^\circ$. The electron (Mass m) touches the sphere again after time 't', find t (Neglect gravity)



(1) $\sqrt{\frac{6r\varepsilon_0 m}{e\rho a}}$

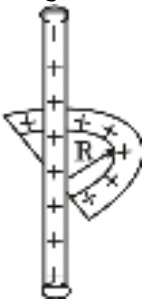
(2) $\sqrt{\frac{6\sqrt{2}r\varepsilon_0 m}{e\rho a}}$

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(3) $\sqrt{\frac{3r\varepsilon_0 m}{e\rho a}}$

(4) $\sqrt{\frac{2r\varepsilon_0 m}{e\rho a}}$

10) Find the force experienced by the semicircular rod charged with a charge q, placed as shown in figure. Radius of the wire is R and the line of charge with linear charge density is passing through its centre and perpendicular to the plane of wire.



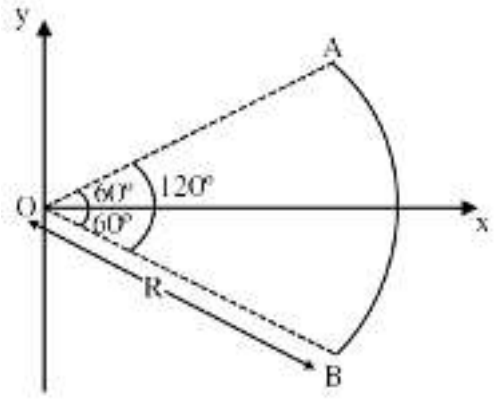
(1) $\frac{\lambda q}{2\pi^2\varepsilon_0 R}$

(2) $\frac{\lambda q}{\pi^2\varepsilon_0 R}$

(3) $\frac{\lambda q}{4\pi^2\varepsilon_0 R}$

(4) $\frac{\lambda q}{4\pi\varepsilon_0 R}$

11) Figure shows a rod AB, which is bent in a 120º circular arc of radius R. A charge (–Q) is uniformly distributed over rod AB. What is the electric field $\vec{E}$ at the centre of curvature O ?



(1) $\frac{3\sqrt{3}Q}{8\pi\varepsilon_0 R^2}(\hat{i})$

(2) $\frac{3\sqrt{3}Q}{8\pi^2\varepsilon_0 R^2}(\hat{i})$

(3) $\frac{3\sqrt{3}Q}{16\pi^2\varepsilon_0 R^2}(\hat{i})$

(4) $\frac{3\sqrt{3}Q}{8\pi^2\varepsilon_0 R^2}(-\hat{i})$

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12) Three identical positive charges Q are arranged at the vertices of an equilateral triangle. The side of the triangle is a. Find the intensity of the field at the vertex of a regular tetrahedron of which the triangle is the base $\left[k = \frac{1}{4\pi\epsilon_0}\right]$

(1) $\sqrt{6}\frac{KQ}{a^2}$

(2) $\sqrt{2}\frac{KQ}{a^2}$

(3) $\sqrt{3}\frac{KQ}{a^2}$

(4) None of these

13) Four point charges –q, +q, +q and –q are placed on y-axis at y = –2d, y = –d, y = +d and y = +2d, respectively. The magnitude of the electric field E at a point on the x-axis at x = D, with D >> d, will behave as :-

(1) $E \propto \frac{1}{D}$

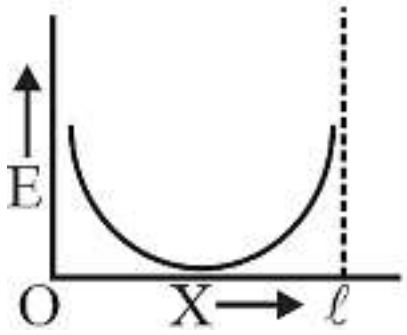
(2) $E \propto \frac{1}{D^3}$

(3) $E \propto \frac{1}{D^2}$

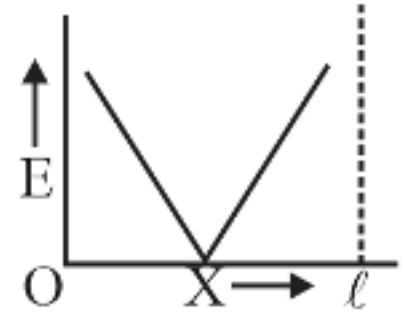
(4) $E \propto \frac{1}{D^4}$

14) Two identical point charges are placed at a separation of ℓ. P is a point on the line joining the charges, at a distance x from any one charge. The field at P is E. E is plotted against x for values of x from close to zero to slightly less than ℓ. Which of the following best represents the resulting curve ?

(1)

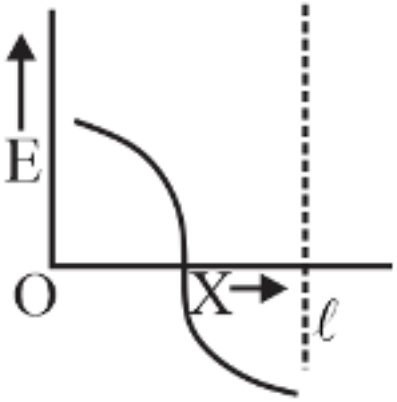


(2)

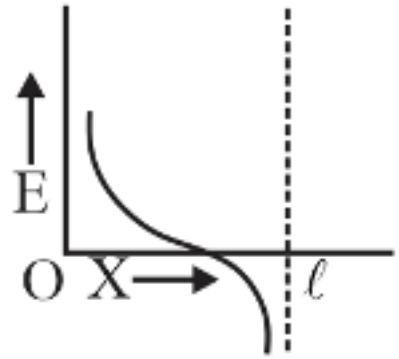


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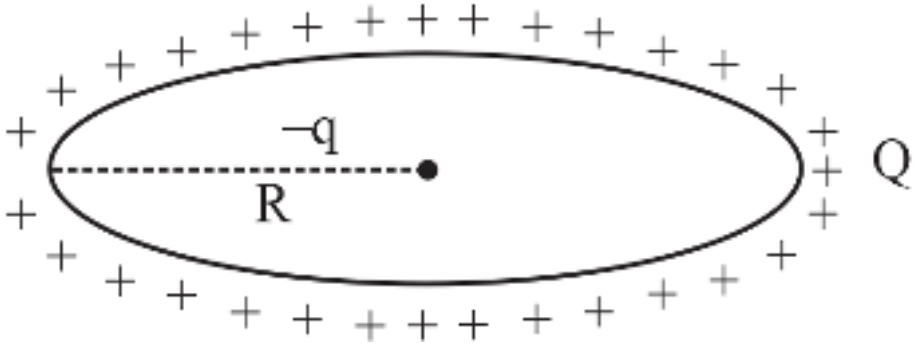
(3)



(4)



15) Figure shows a particle of mass m and charge –q at the centre of a fixed uniformly charged ring in a gravity free zone. The angular frequency of small oscillations of the particle along the axis of the ring is



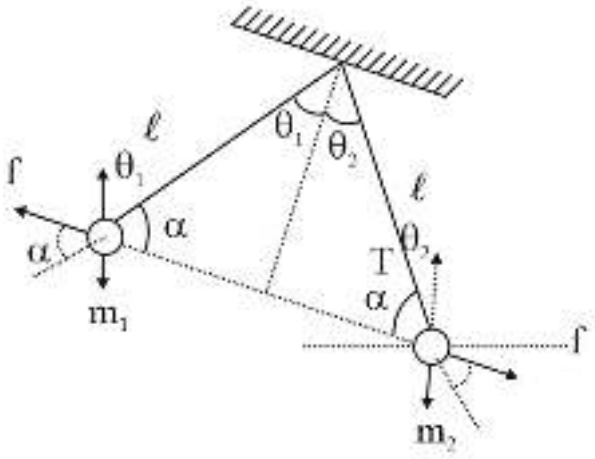
(1) $\sqrt{\frac{qQ}{2\pi\varepsilon_0 mR^3}}$

(2) $\sqrt{\frac{qQ}{4\pi\varepsilon_0 mR^3}}$

(3) $\sqrt{\frac{2qQ}{\pi\varepsilon_0 mR^3}}$

(4) $\sqrt{\frac{3}{2}\frac{qQ}{\pi\varepsilon_0 mR^3}}$

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16) $\alpha = 90\dfrac{(\theta_1 + \theta_2)}{2}$

on system

$T\cos\theta_1 + T\cos\theta_2 = mg + mg$

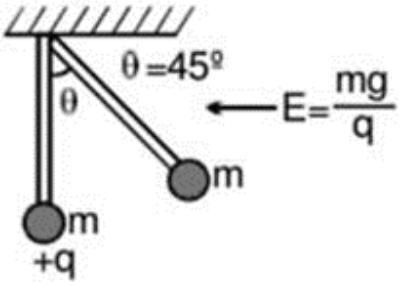
(1) $\dfrac{m_1}{m_2} = \dfrac{\sin\theta_2}{\sin\theta_1}$

(2) $\dfrac{q_1}{q_2} = \dfrac{\sin\theta_1}{\sin\theta_2}$

(3) $\dfrac{m_1}{m_2} = \dfrac{\sin\theta_2}{\sin\theta_1}$

(4) $\dfrac{m_1}{m_2} = \dfrac{\tan\theta_2}{\tan\theta_1}$

17) In space of horizontal $EF\left(E = \dfrac{(mg)}{q}\right)$ exist as shown in figure and a mass m attached at the end of a light rod. If mass m is released from the position shown in figure find the angular velocity of the rod when it passes through the bottom most position.



(1) $\sqrt{\dfrac{g}{\ell}}$

(2) $\sqrt{\dfrac{2g}{\ell}}$

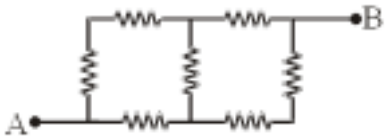
(3) $\sqrt{\dfrac{3g}{\ell}}$

(4) $\sqrt{\dfrac{5g}{\ell}}$

18) Seven resistors each of resistance 5Ω are connected as shown in the figure. The equivalent

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resistance between the points A and B is :-



(1) 5Ω

(2) 5.5Ω

(3) 7Ω

(4) 7.5Ω

19) An electric tea kettle has two heating coils. When one the coils is switched on, the tea begins to boil in 6 minutes. When the other is switched on, the boiling begins in 8 minutes. In what time will the boiling begin if both coils are switched on simultaneously (i) in series
(ii) in parallel ?

(1) 14 min, 3.43 min

(2) 12 min, 3.43 min

(3) 3.43 min, 12 min

(4) 3.43 min, 14 min

20) Two electric bulbs marked 25W - 220 V and 100W - 220 V are connected in series to a 440 V supply. Which of the bulbs will fuse ?

(1) 100 W

(2) 25 W

(3) Neither

(4) Both

21) Two cells of same emf but different internal resistances r_1 and r_2 are connected in series with a resistance R. The value of resistance R, for which the potential difference across second cell is zero, is

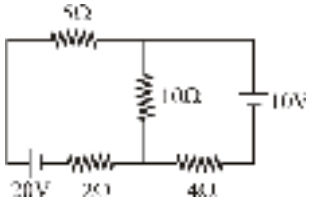
(1) $r_2 - r_1$

(2) $r_1 - r_2$

(3) r_1

(4) r_2

22) In the figure shown, the current in the 10 V battery is close to :



(1) 0.36 A from negative to positive terminal.

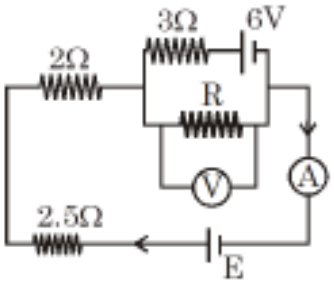
(2) 0.71 A from positive to negative terminal

(3) 0.21 A from positive to negative terminal.

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(4) 0.42 A from positive to negative terminal.

23) In a circuit, voltmeter reads 3V and the ammeter reads 2A.Then wrong option is :-



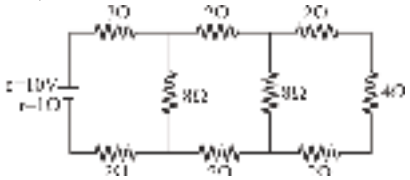
(1) the resistance R is 1 ohm

(2) the emf E is 12 V

(3) the current in 3W resistor is 1A

(4) the emf E is 9V

24) In the circuit shown, the cell has emf = 10V and internal resistance = 1Ω :–



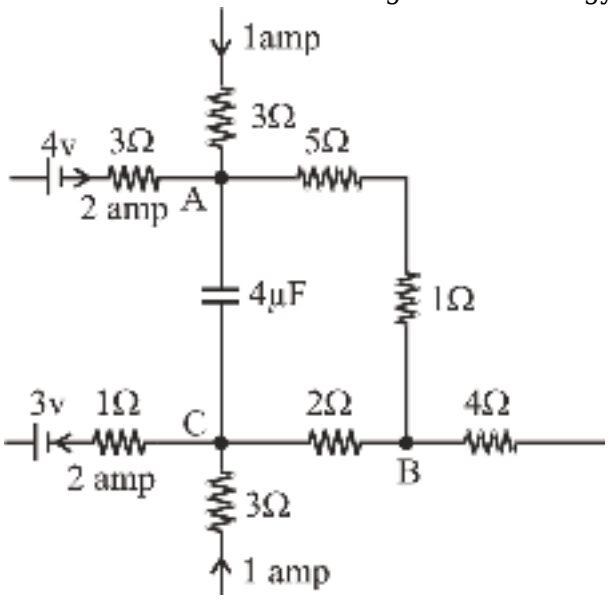
(1) The current through the 3Ω resistor is 2A

(2) The current through the 3Ω resistor is 0.5A

(3) The current through the 4Ω resistor is 0.5A

(4) The current through the 4Ω resistor is 0.25A

25) A part of the circuit in a steady along with the current flowing in the branches, the value of resistance etc. is shown in figure. The energy strored in capacitor (4μF) is :-



(1) 1.6 J

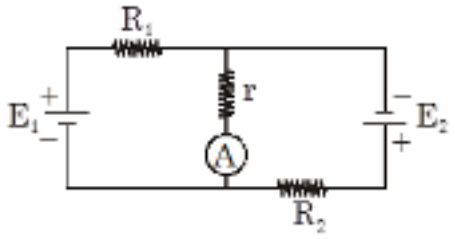
(2) 0.8 J

(3) 1600 μF

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(4) 800 µJ

26) If the reading of ideal ammeter A in circuit shown here is zero, then the ratio $\frac{E_1}{E_2}$ of the emf's of



the two batteries must be :-

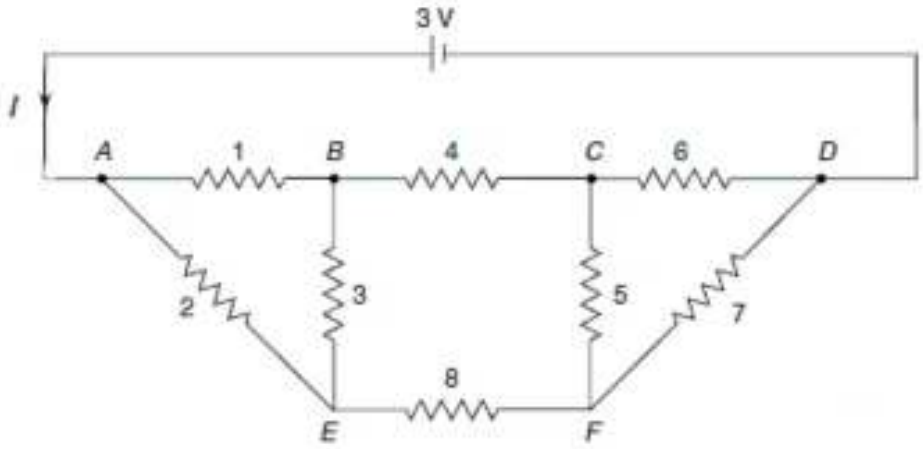
(1) 1

(2) $\frac{R_1}{R_2}$

(3) $\frac{R_2}{R_1}$

(4) None of these

27) Figure shows a network of eight resistors numbered 1 to 8 each equal to 2Ω connected to a 3v battery of negligible internal resistance. The current i in the circuit is



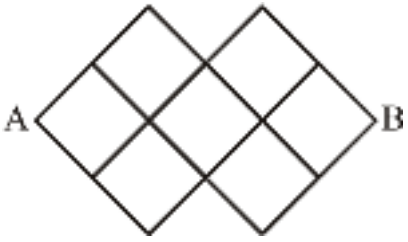
(1) 0.25 A

(2) 0.5 A

(3) 0.75 A

(4) 1.0 A

28) In the shown wire frame, each side of a square (the smallest square) has a resistance $\frac{R}{2}$. The



equivalent resistance of the circuit between the points A and B is :-

(1) $\frac{R}{2}$

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(2) R

(3) 2R

(4) 4R

29) Incandescent bulbs are designed by keeping in mind that the resistance of their filament increases with the increase in temperature. If at room temperature, 100 W, 60 W and 40 W bulbs have filament resistance R_{100}, R_{60} and R_{40}, respectively, the relation between these resistance is

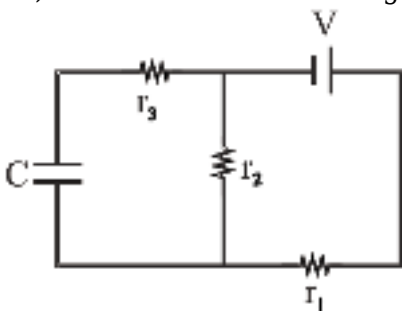
(1) $\frac{1}{R_{100}} = \frac{1}{R_{40}} + \frac{1}{R_{60}}$

(2) $R_{100} = R_{40} + R_{60}$

(3) $R_{100} > R_{60} > R_{40}$

(4) $\frac{1}{R_{100}} > \frac{1}{R_{60}} > \frac{1}{R_{40}}$

30) In the circuit shown in figure, the final voltage drop across the capacitor C is :-



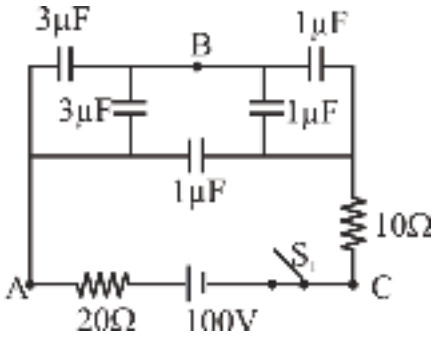
(1) $\frac{Vr_1}{r_1 + r_2}$

(2) $\frac{Vr_2}{r_1 + r_2}$

(3) $\frac{V(r_1 + r_2)}{r_2}$

(4) $\frac{V(r_2 + r_1)}{r_1 + r_2 + r_3}$

31) $V_A - V_B$ after a long time of closing of switch is :-



(1) 25 volt

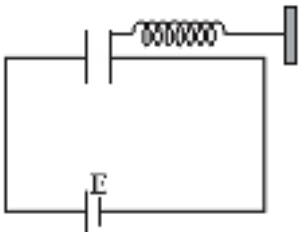
(2) 75 volt

(3) 50 volt

(4) 100 vlot

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32) One plate of a capacitor is fixed and the other is connected to a spring as shown in figure. Area of both the plates is A. In steady state (equilibrium), separation between the plates is 0.8d (spring was unstretched and the distance between the plates was d when the capacitor was uncharged).



The force constant of the spring is approximately.

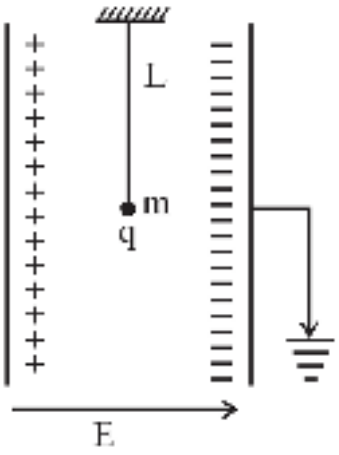
(1) $\frac{125}{32}\frac{\varepsilon_0 AE^2}{d^3}$

(2) $\frac{2\varepsilon_0 AE^2}{d^3}$

(3) $\frac{6\varepsilon_0 E^2}{Ad^2}$

(4) $\frac{\varepsilon_0 AE^3}{2d^3}$

33) A simple pendulum of length L is placed between the plates of a parallel plate capacitor having electric field E, as shown in figure. Its bob has mass m and charge q. The time period of the



pendulum is given by

(1) $2\pi\sqrt{\frac{L}{\sqrt{g^2+\left(\frac{qE}{m}\right)^2}}}$

(2) $2\pi\sqrt{\frac{L}{\sqrt{g^2-\frac{q^2E^2}{m^2}}}}$

(3) $2\pi\sqrt{\frac{L}{\left(g+\frac{qE}{m}\right)}}$

(4) $2\pi\sqrt{\frac{L}{\left(g-\frac{qE}{m}\right)}}$

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34) A parallel plate capacitor with plate area 'A' and distance of separation 'd' is filled with a dielectric. What is the capacity of the capacitor when permittivity of the dielectric varies as :

$\varepsilon(x) = \varepsilon_0 + kx$, for $\left(0 < x \leqslant \frac{d}{2}\right)$

$\varepsilon(x) = \varepsilon_0 + k(d-x)$, for $\left(\frac{d}{2} \leqslant x \leqslant d\right)$

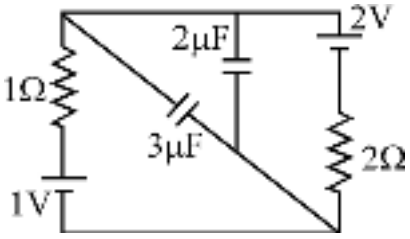
(1) $\left(\varepsilon_0 + \frac{kd}{2}\right)^{2/kA}$

(2) $\frac{kA}{2\ln\left(\frac{2\varepsilon_0+kd}{2\varepsilon_0}\right)}$

(3) 0

(4) $\frac{kA}{2}\ln\left(\frac{2\varepsilon_0}{2\varepsilon_0-kd}\right)$

35) In the circuit shown, the charge on the 3μF capacitor at steady state will be



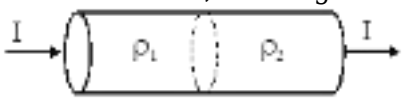
(1) 6 μC

(2) 4 μC

(3) $\frac{2}{3}$ μC

(4) 3 μC

36) Two long straight cylindrical conductors with resistivities ρ_1 and ρ_2 respectively are joined together as shown in figure. If current I flows through the conductors, the magnitude of the total free charge at the interface of the two conductors is :-



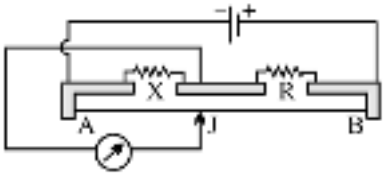
(1) zero

(2) $\frac{(\rho_1-\rho_2)\,I\varepsilon_0}{2}$

(3) $\varepsilon_0 I\,|\rho_1-\rho_2|$

(4) $\varepsilon_0 I\,|\rho_1+\rho_2|$

37) The figure shows a metre-bridge circuit, with $AB = 100\ cm$, $X = 12\Omega$ and $R = 18\Omega$, and the jockey J in the position of balance. If R is now made 8Ω, through what distance will J have to be moved to obtain balance?



Watermarkly

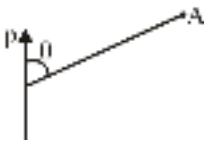
(1) 10 *cm*

(2) 20 *cm*

(3) 30 *cm*

(4) 40 *cm*

38) The electric field at A due to dipole p is perpendicular to p. The angle θ is :-



(1) 0°

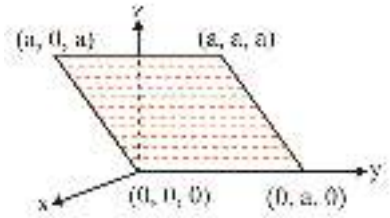
(2) 90°

(3) $\tan^{-1} 2$

(4) $\frac{\tan^{-1}}{\sqrt{2}}$

39)

Consider an electric field $\vec{E} = E_0\hat{x}$, where E_0 is a constant. The flux through the shaded area (as shown in the figure) due to this field is:-



(1) $2E_0a^2$

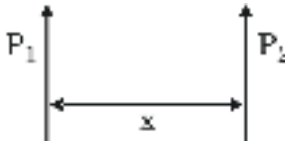
(2) $\sqrt{2}E_0a^2$

(3) E_0a^2

(4) $\frac{E_0a^2}{\sqrt{2}}$

40)

Two very small dipoles are kept as shown in figure.



The interactional potential energy of two dipole is

(1) $\frac{1}{2\pi\varepsilon_0}\left(\frac{P_1P_2}{x^3}\right)$

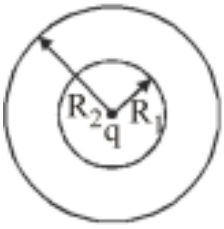
(2) $\frac{1}{4\pi\varepsilon_0}\left(\frac{P_1P_2}{x^3}\right)$

(3) $\frac{3}{4\pi\varepsilon_0}\left(\frac{P_1P_2}{x^3}\right)$

Watermarkly

(4) Zero

41) The region between two concentric spheres of radii R_1 and $R_2(R_1 < R_2)$ has volume charge density $\rho = \frac{b}{r}$, where b is constant and r is the radial distance. A point charge q = 16 μC is placed at the origin, r = 0. Find the value of b (in SI units) for which the electric field in the region between spheres is constant.



(Take : $R_2 = 2R_1 = \frac{4}{\sqrt{\pi}}$mm)

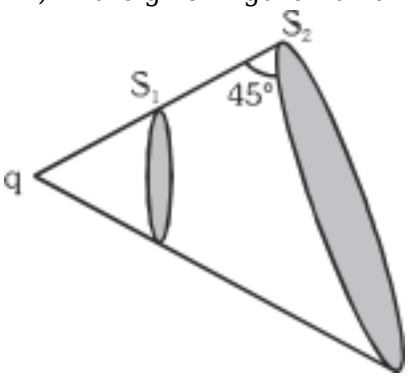
(1) 1

(2) 2

(3) 4

(4) 8

42) In the given figure flux through surface S_1 is ϕ_1 & through S_2 is ϕ_2. Which is correct ?



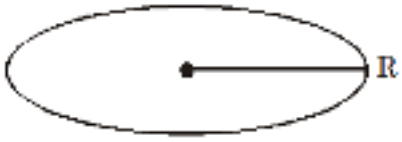
(1) $\phi_1 = \phi_2$

(2) $\phi_1 > \phi_2$

(3) $\phi_1 < \phi_2$

(4) None of these

43) A circular disc of radius R carries surface charge density $\sigma(r) = \sigma^0\left(1 - \frac{r}{R}\right)$, where σ^0 is a constant and r is the distance from the center of the disc. Assume that the disc is in xz plane and its axis is y axis. We have a sphere of radius R and centre at $\left(0, \frac{\sqrt{3}R}{2}, 0\right)$. What is the flux of electric field through this sphere?



(1) $\pi\sigma_0\frac{R^2}{2\epsilon_0}$

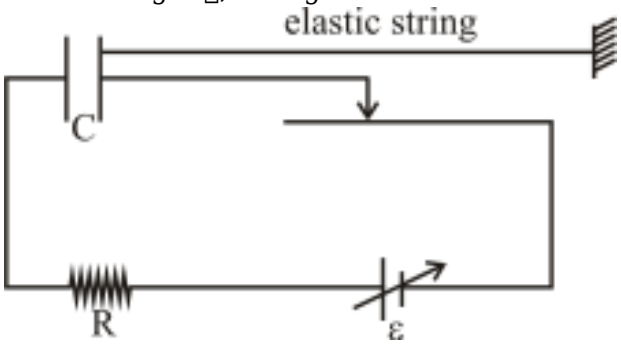
Watermarkly

(2) $2\pi\sigma_0\frac{R^2}{3\epsilon_0}$

(3) $\pi\sigma_0\frac{R^2}{6\epsilon_0}$

(4) $\pi\sigma_0\frac{R^2}{12\epsilon_0}$

44) Consider a circuit shown in the figure. The emf of the battery is slowly increased to a final value of E_0 from zero. One plate of the capacitor is fixed while the other plate is tied to an elastic string of natural length □, Young's modulus Y & area of cross-section A :-



(1) Capacitance of the capacitor continuously decreases as time proceeds

(2) Maximum value of elongation in the string is less than 'd' seperation between the plates initially for any value of E_0.

(3) Energy stored in the capacitor is more as compared to the case when the plates of capacitor were kept fixed.

(4) Extention in string increases linearly with value of emf of battery.

45) A charge of 1 μC is placed at point P(3,2,5). Find the electric field vector at point S(0,–4,3) :-

(1) $\frac{-9}{343}\left(3\hat{i}+6\hat{j}+2\hat{k}\right)\times10^3\text{N/c}$

(2) $\frac{9}{343}\left(3\hat{i}-6\hat{j}+\hat{k}\right)\times10^3\text{N/c}$

(3) $\frac{-3}{343}\left(3\hat{i}+6\hat{j}+2\hat{k}\right)\times10^3\text{N/c}$

(4) $\frac{9}{343}\left(3\hat{i}+6\hat{j}+2\hat{k}\right)\times10^3\text{N/c}$

CHEMISTRY

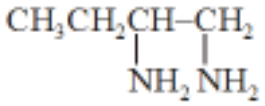
$$CH_3CH_2\underset{Br}{CH}-\underset{Br}{CH_2} \xrightarrow[\text{(ii) } NaNH_2 \text{ in liq } NH_3]{\text{(i) KOH alc.}}$$

1) The major product of the following reaction is:

(1) $CH_3CH_2C\equiv CH$

Watermarkly

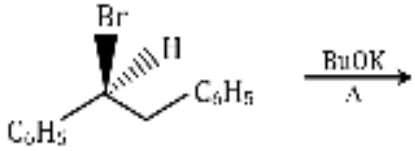
NOVA @examlix

(2) 

(3) $CH_3CH=C=CH_2$

(4) $CH_3CH=CHCH_2NH_2$

2) The major product obtained in the following reaction is :-



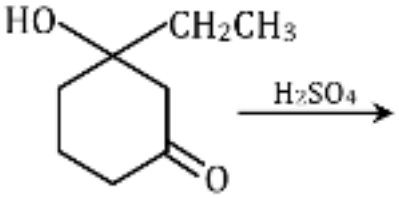
(1) $(\pm)C_6H_5CH(O\text{-}tBu)CH_2C_6H_5$

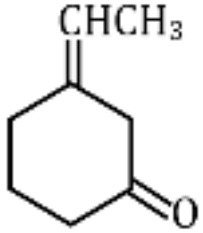
(2) $C_6H_5CH=CHC_6H_5$

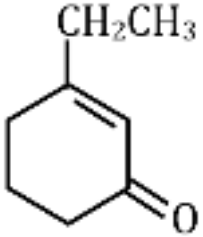
(3) $(+)C_6H_5CH(O\text{-}tBu)CH_2C_6H_5$

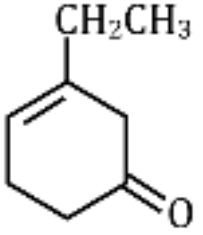
(4) $(-)C_6H_5CH(O\text{-}tBu)CH_2C_6H_5$

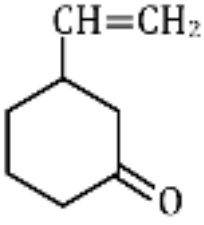
3) The major product of the following reaction is:



(1) 

(2) 

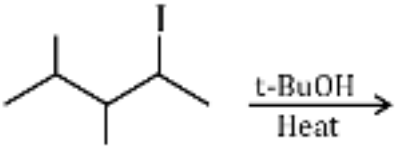
(3) 

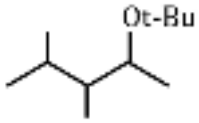
(4) 

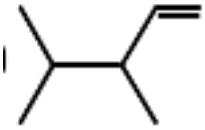
Watermarkly

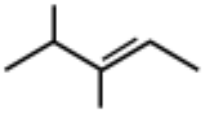
@examlix

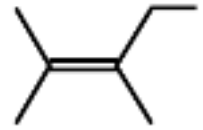
4) The major product in the following reaction is :

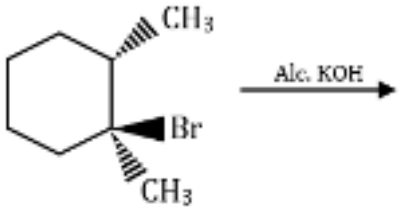


(1) 

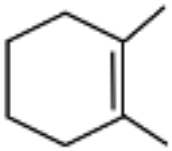
(2) 

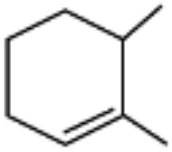
(3) 

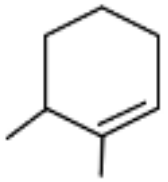
(4) 

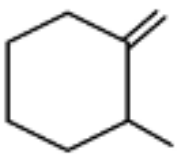
5) 

is subjected to E2. The product will be

(1) 

(2) 

(3) 

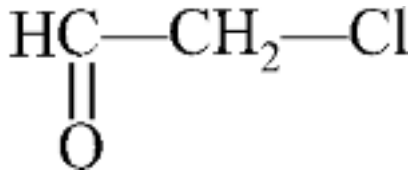
(4) 

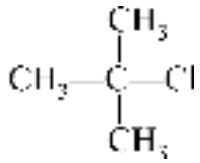
6) Which of the following is most reactive towards S_N2 mechanism

(1) CH_3–Cl

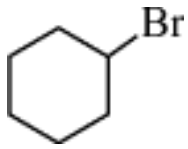
(2) CH_2=CH–CH_2–Cl

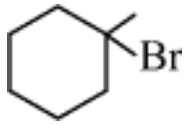
Watermarkly

(3) 

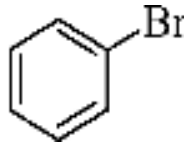
(4) 

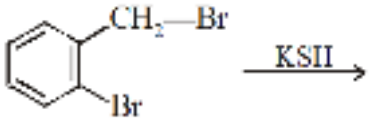
7) Which among the following will show highest rate of S_N1 on hydrolysis

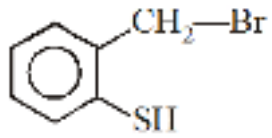
(1) 

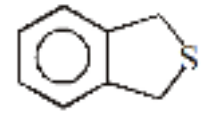
(2) 

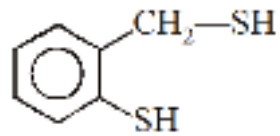
(3) 

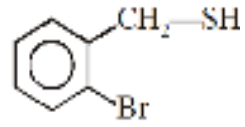
(4) 

8) Find the major product of the following reaction : 

(1) 

(2) 

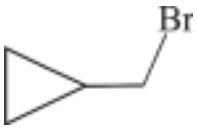
(3) 

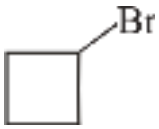
(4) 

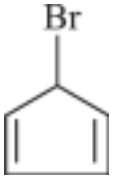
9) Which of the following can not give S_N1 reaction easily?

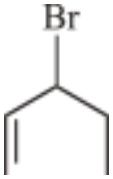
Watermarkly

@examlix

(1) 

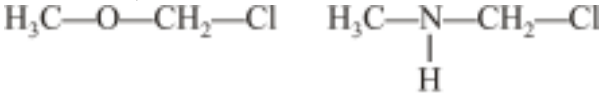
(2) 

(3) 

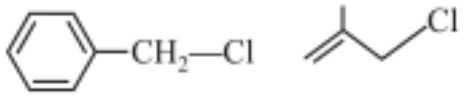
(4) 

10)

Rate of S_N1 reaction is :



(P) (Q)



(R) (S)

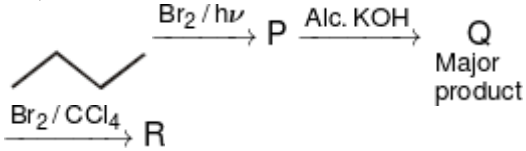
(1) P > Q > R > S

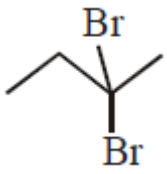
(2) R > Q > P > Q

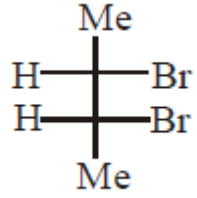
(3) Q > P > R > S

(4) Q > P > S > R

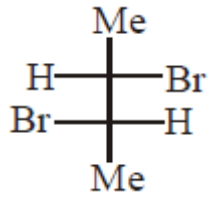
11) Predict the structure of R from the given sequence of reaction :

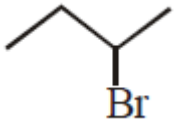


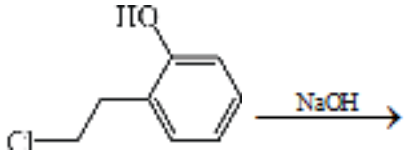
(1) 

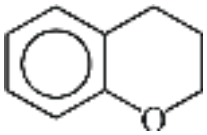
(2) 

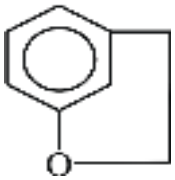
Watermarkly

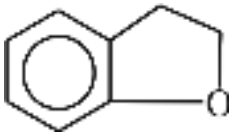
(3) 

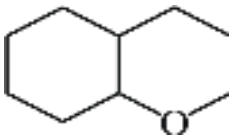
(4) 

12)  product (P). , P is -

(1) 

(2) 

(3) 

(4) 

13) Select suitable reason for non-occurrence of the following reaction ?

$$Br^{\ominus} + CH_3OH \longrightarrow CH_3Br + OH^{\ominus}$$

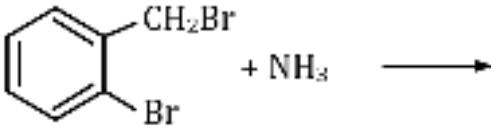
(1) attacking nucleophile is stronger one

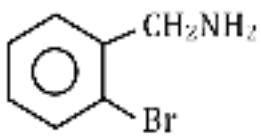
(2) leaving group is strong base than nucleophile

(3) Alcohols are not good substrate for S_N reaction.

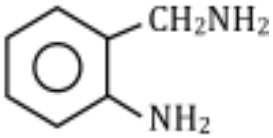
(4) Hydroxide ions are weak bases

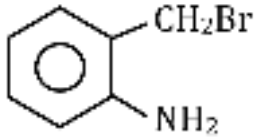
14) The major product in the given reaction:



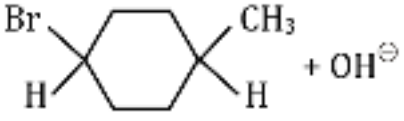
(1) 

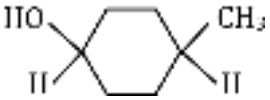
Watermarkly

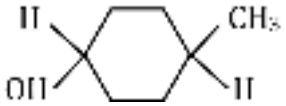
(2) 

(3) 

(4) All of these

15)  A ; A is :

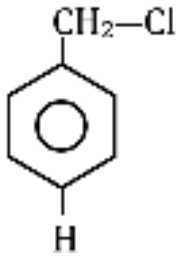
(1) 

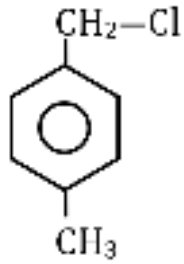
(2) 

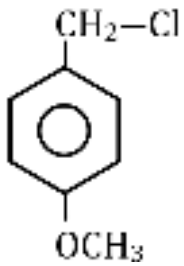
(3) Both

(4) None

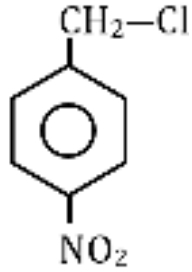
16) Which of the following is most reactive toward S_N2.

(1) 

(2) 

(3) 

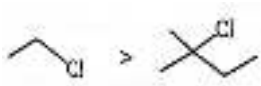
Watermarkly

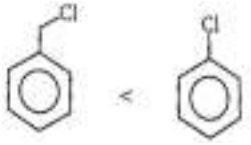
(4) 

17) The correct order of rate of S_N1 is:

(1) 

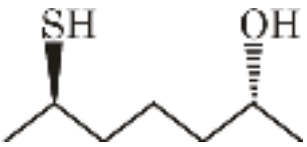
(2) 

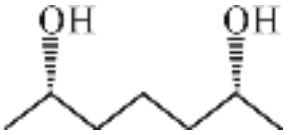
(3) 

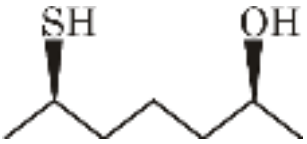
(4) 

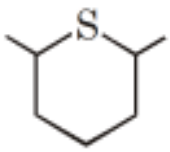
18) 

Major product of above reaction is :

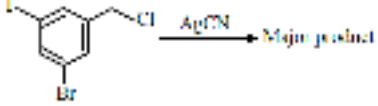
(1) 

(2) 

(3) 

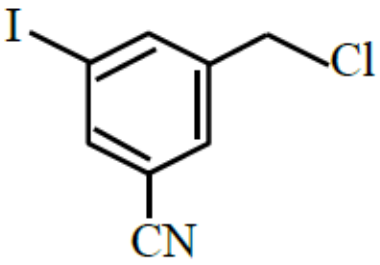
(4) 

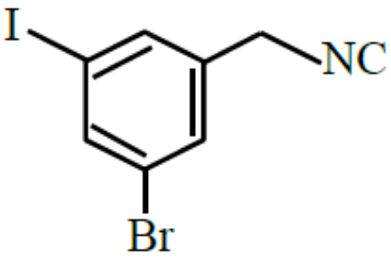
19) The structure of the major product formed in the following reaction is :

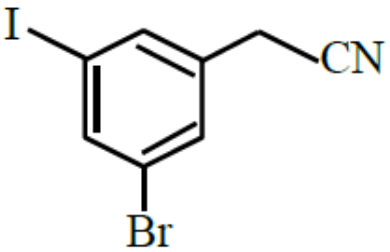


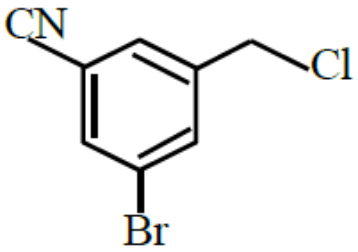
Watermarkly

@examlix

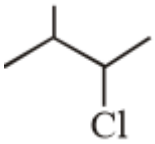
(1) 

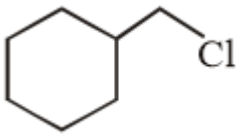
(2) 

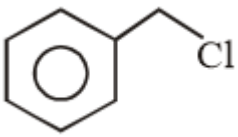
(3) 

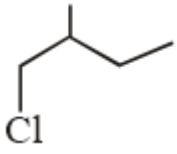
(4) 

20) Compounds which give same major product by S_N^1 & S_N^2 mechanism -

(1) 

(2) 

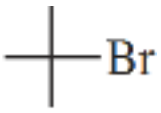
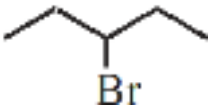
(3) 

(4) 

21) Correct order of E_2 / S_N2 ratio is :

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P ⇒  Q ⇒ 

R ⇒ CH_3 – Br S ⇒ CH_3 – CH_2 – Br

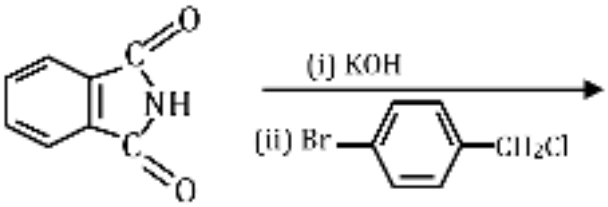
(1) P > Q > S > R

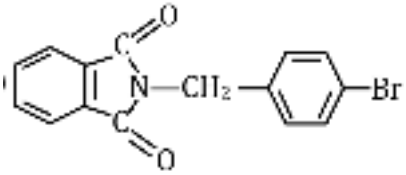
(2) P > Q > R > S

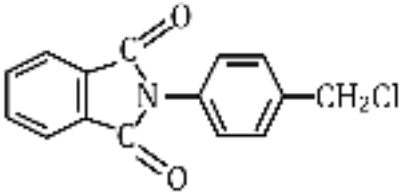
(3) R > S > Q > P

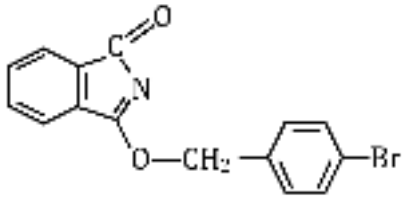
(4) P > S > Q > R

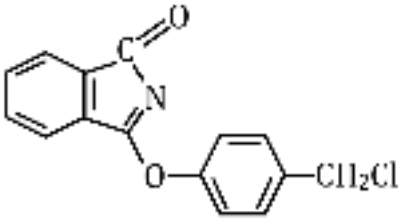
22) The major product of the following reaction is



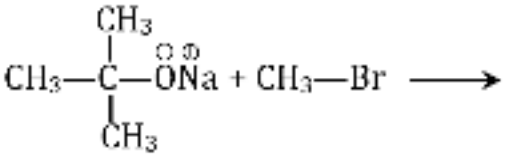
(1) 

(2) 

(3) 

(4) 

23) Which reaction gives elimination as a major product

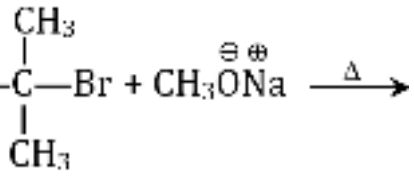
(1) 

(2) CH_3—CH_2—Br + NaCN →

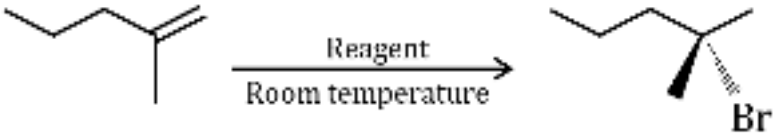
(3) CH_3—CH_2—Br + NaI →

Watermarkly

@examlix

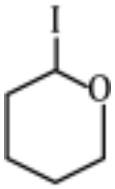
(4) 

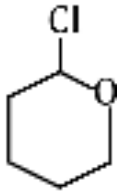
24) What is the reagent(s) needed for the following reaction?

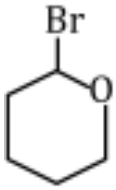


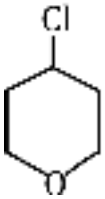
(1) HCl
(2) NaBr
(3) HBr
(4) Br_2

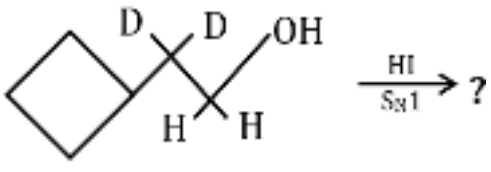
25) Which one of the following compounds will be most reactive for S_N1 reactions?

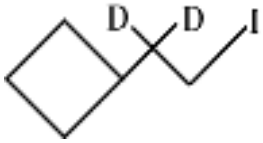
(1) 

(2) 

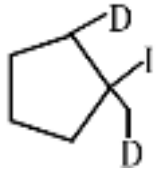
(3) 

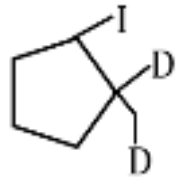
(4) 

26) 

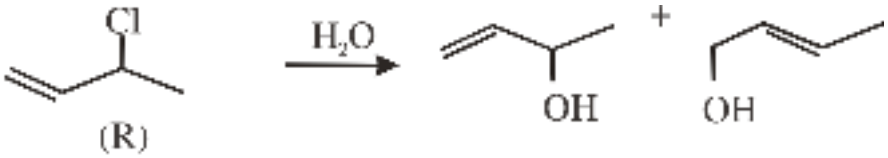
(1) 

Watermarkly

(2) 

(3) 

(4) None of these



(R)

27) optically active

What is correct for the above reaction:

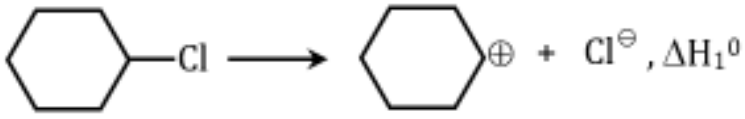
(1) Total three products are formed

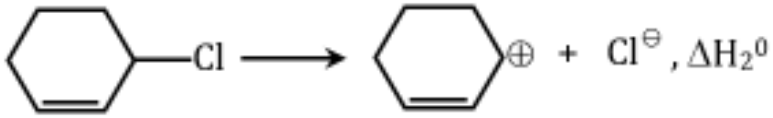
(2) Products mixture is optically active

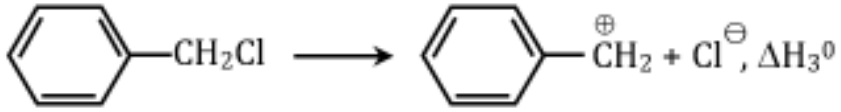
(3) Total two products are Chiral

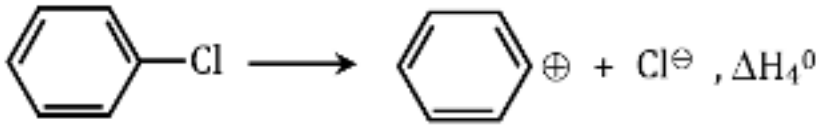
(4) All of the above

28) For the reactions

(I) 

(II) 

(III) 

(IV) 

The correct decreasing order of enthalpies of reaction for producing carbocation is :

(1) $\Delta H_1^\circ > \Delta H_2^\circ > \Delta H_3^\circ > \Delta H_4^\circ$

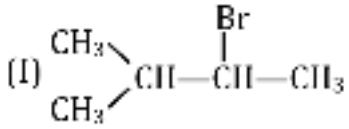
(2) $\Delta H_4^\circ > \Delta H_1^\circ > \Delta H_2^\circ > \Delta H_3^\circ$

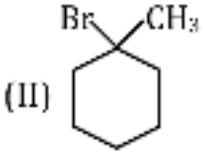
(3) $\Delta H_3^\circ > \Delta H_2^\circ > \Delta H_1^\circ > \Delta H_4^\circ$

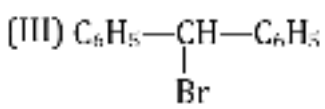
(4) $\Delta H_2^\circ > \Delta H_1^\circ > \Delta H_4^\circ > \Delta H_3^\circ$

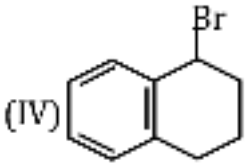
29) Consider the S_N1 solvolysis of the following halides in aqueous formic acid:

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(I) 

(II) 

(III) 

(IV) 

Decide decreasing order of reactivity of above alkyl halide?

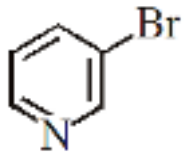
(1) III > IV > II > I

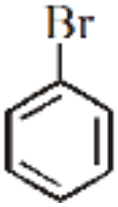
(2) II > IV > I > III

(3) I > II > III > IV

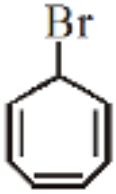
(4) III > I > II > IV

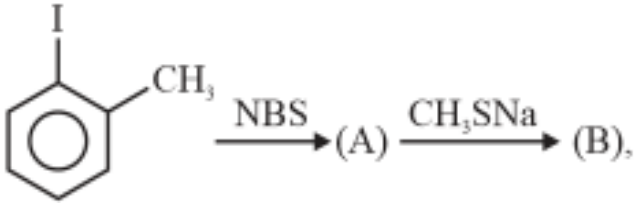
30) Which of the following compounds will produce a precipitate with $AgNO_3$?

(1) 

(2) 

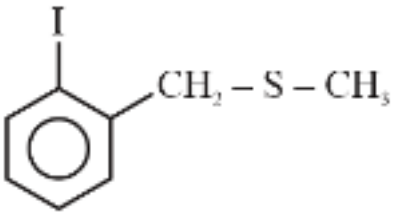
(3) 

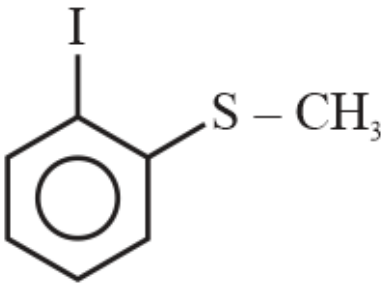
(4) 

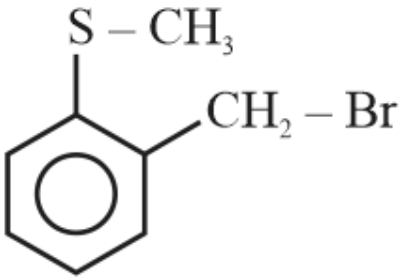
31) 

The product (B) is

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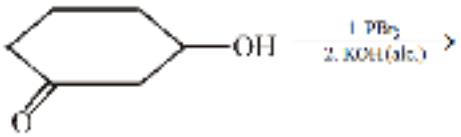
(1) 

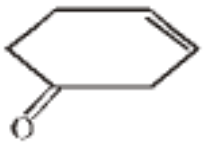
(2) 

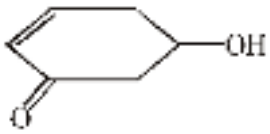
(3) 

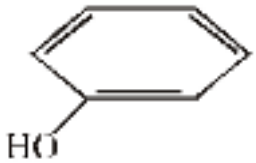
(4) None of these

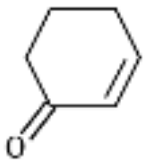
32) The mojor product of the following reaction is :



(1) 

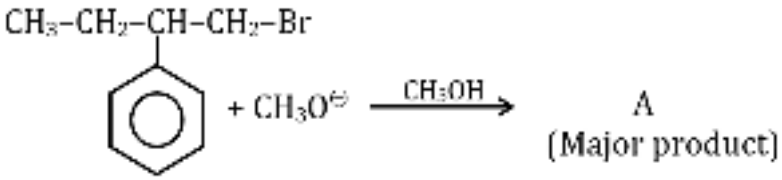
(2) 

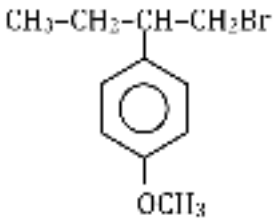
(3) 

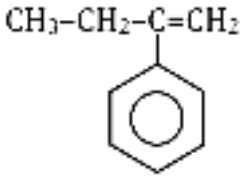
(4) 

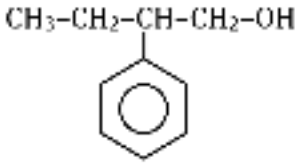
33) The major product (A) formed in the reaction given below is :

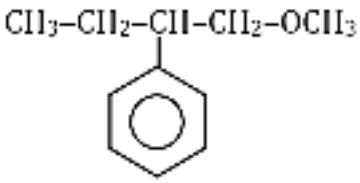
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(1) 

(2) 

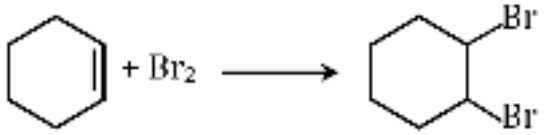
(3) 

(4) 

34) For the following reactions :-

(a) $CH_3CH_2CH_2Br + KOH \rightarrow CH_3CH=CH_2 + KBr + H_2O$

(b) 

(c) 

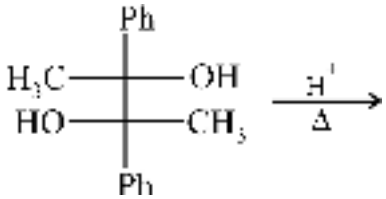
Which of the following statement is correct?

(1) (a) and (b) are elimination reaction and (c) is addition reaction

(2) (a) is elimination, (b) is substitution and (c) is addition reaction

(3) (a) is elimination, (b) and (c) are substitution reactions

(4) (a) is substitution, (b) and (c) are addition reaction

35) 

Number of pinacolone obtained as a major product during above reaction is :

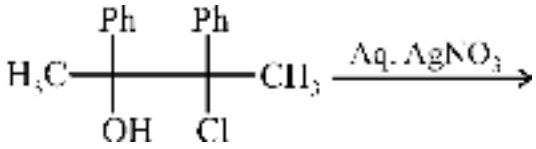
Watermarkly

(1) 3

(2) 4

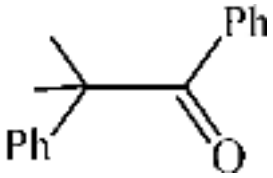
(3) 2

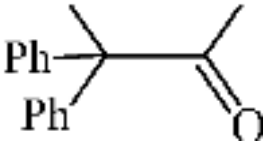
(4) 1

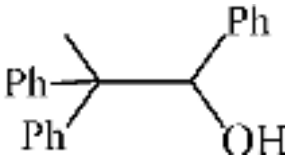


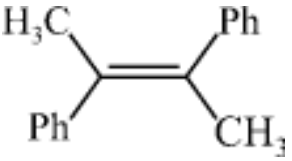
36) Major Product

The major product is :

(1) 

(2) 

(3) 

(4) 

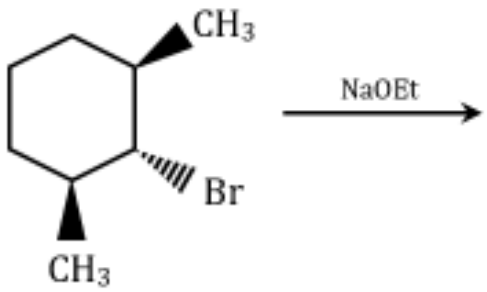
37) Which product is formed in E1 reaction of 1 – Bromo -2, 3 – dimethylcyclohexane?

(1) 3, 4 – Dimethylcyclohexane

(2) 2, 3 – Dimethylcyclohexane

(3) 4, 5 – Dimethylcyclohexene

(4) 1, 2 – Dimethylcylohexene

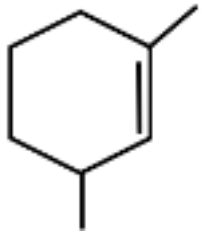


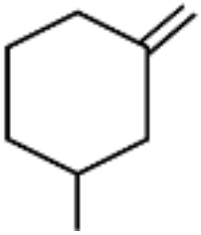
38)

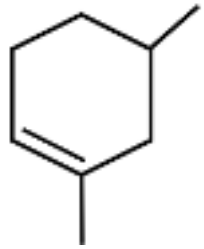
is subjected to E2 reaction major product will be

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@examlix

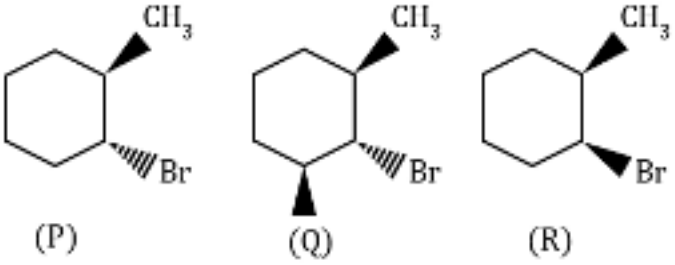
(1) 

(2) 

(3) 

(4) None of these

39) Arrange the following in order of decreasing rate of $\mathbf{E_2}$ reaction ___?



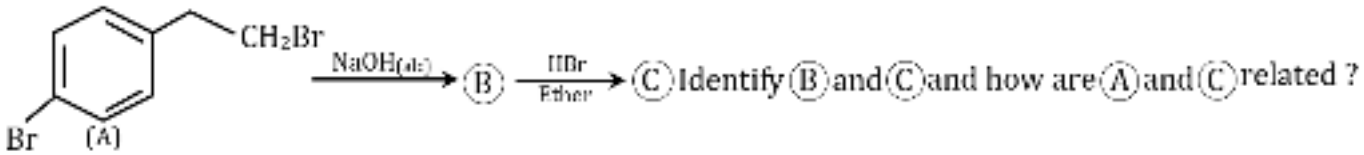
(1) P > Q > R

(2) R > P > Q

(3) R > Q > P

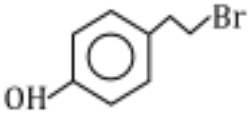
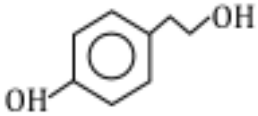
(4) Q > P > R

40)

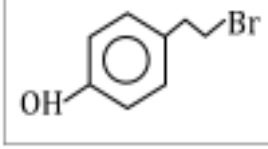
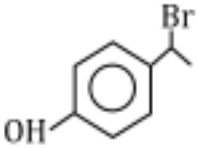
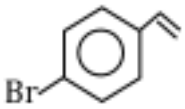
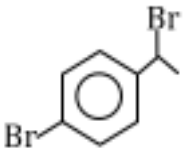
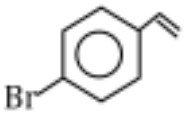
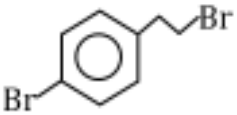


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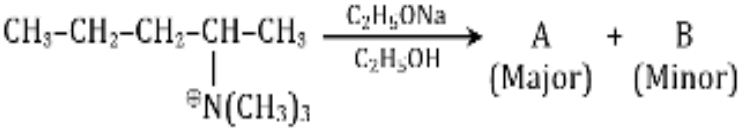
(1)

		functional group isomers

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(2)			Derivative
(3)			position isomers
(4)			chain isomers

41) Identify the correct statement for the below given transformation.



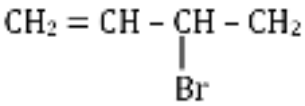
(1) A - $CH_3CH_2CH = CH\text{-}CH_3$, B – $CH_3CH_2CH_2CH = CH_2$, Saytzeff products

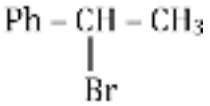
(2) A - $CH_3CH_2CH = CH\text{-}CH_3$, B – $CH_3CH_2CH_2CH = CH_2$, Hofmann products

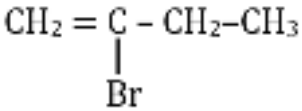
(3) A - $CH_3CH_2CH_2CH = CH_2$, B – $CH_3CH_2CH = CHCH_3$, Hofmann products

(4) A - $CH_3CH_2CH_2CH = CH_2$, B – $CH_3CH_2CH = CHCH_3$, Saytzeff products

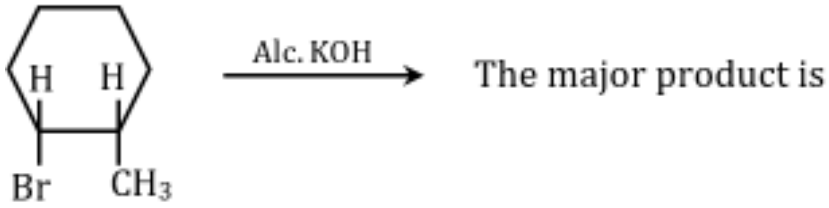
42) Which alkyl halide would you expect to be least reactive in E1 reaction ?

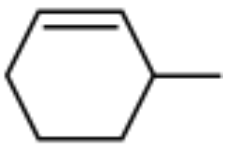
(1) 

(2) 

(3) 

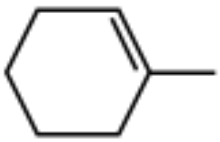
(4) Ph – CH_2 – CH_2 Br

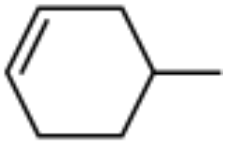
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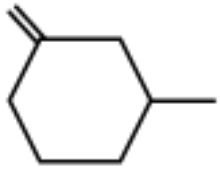
(1) 

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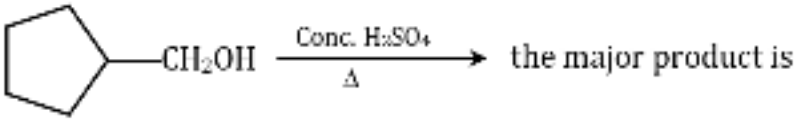
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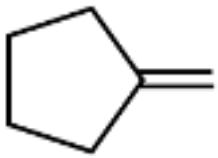
(2) 

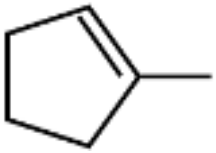
(3) 

(4) 

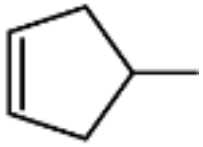
44) For the reaction,

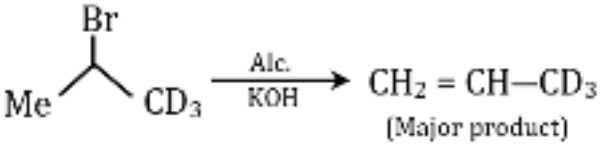


(1) 

(2) 

(3) 

(4) 

45) 

By which mechanism does the above reaction proceed?

(1) E1

(2) E2

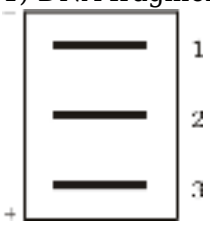
(3) E1 cB

(4) γ-Elimination

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BIOLOGY

1) DNA fragments separated by gel electrophoresis are shown. Mark the correct statement :-



(1) Band '3' contains more positively charged DNA molecule than '1'

(2) Band '3' has longer DNA fragment than '1' and '2'

(3) Band '1' has longer DNA fragment than '2' and '3'

(4) All the bands have equal length and charges but differ in base composition

2) Read and following statements carefully and select the correct option :
(A) Source of the restriction enzyme Hind III is E.coli
(B) In biolistic method of gene transfer, microparticles made up of gold or tungsten are coated with foreign DNA.
(C) Micro-injection method for injecting desired gene is used for animal cell.
(D) Primers are chemically synthesized oligonucleotides that are complementary to the regions of DNA in PCR.
How many of the above statements are correct?

(1) Four

(2) Three

(3) Two

(4) One

3) **How many of the statements given below are correct? Select the appropriate option.**
(a) Genetic engineering involves techniques to directly alter the chemistry of genetic material (DNA/RNA).
(b) Bioprocess engineering deals with maintaining suitable conditions for growth of organisms and product formation.
(c) Biotechnology principles are restricted only to agriculture and not applied in medicine.
(d) Recombinant DNA technology is a major component of genetic engineering.

(1) (a) and (b)

(2) (a), (b) and (d)

(3) (b), (c) and (d)

(4) All four statements

4) Select the correct match :

(1)	Insertional inactivation	-	Differentiate recombinants from non recombinants

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(2)	Recombinant protein	-	Any protein encoding gene is expressed in a homologous host.
(3)	Down stream processing	-	A series of process which start from raw material and to completion of biosynthetic product
(4)	Cloning	-	Introduction of the identified DNA into the host.

(1) 1

(2) 2

(3) 3

(4) 4

5) Read the following statements and choose the correct set of statements :
(a) In micro injection method cells are bombarded with high velocity micro-particles of gold
(b) In PCR taq DNA polymerase denatured during the high temperature
(c) A stirred-tank reactor is usually cylindrical or with a curved base to facilitate the mixing of the reactor contents
(d) Transformation is a procedure through which a piece of DNA is introduced in a host bacterium
Choose the correct answer from the options given below :-

(1) (a),(b) and (d) only

(2) (c) and (d) only

(3) (a),(b),(c) and (d) only

(4) (a) and (b) only

6) In the following question, a statement of assertion (A) is followed by a statement of reason (R).
A : Restriction enzymes link foreign DNA to a plasmid in order to produce rDNA.
R : Restriction enzymes are called molecular glue.
In the light of above statements, select the correct option.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

7) **Assertion (A):** *E.coli* having pBR322 with an insert at *Pvu* I site becomes ampicillin sensitive.
Reason (R): Recognition site for *Pvu* II is present in tet^R gene of pBR322.
In the light of above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true and (R) is false

(4) (A) is false and (R) is true

8) The colonies of recombinant bacteria appear white in contrast to blue colonies of non-recombinant bacteria because of :

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(1) Inactivation of glycosidase enzyme in recombinant bacteria

(2) Non-recombinant bacteria containing beta-galactosidase

(3) Insertional inactivation of alpha-galactosidase in non-recombinant bacteria

(4) Insertional inactivation of beta-galactosidase in recombinant bacteria

9) Why is the "Heat Shock" step necessary in the transformation of *E. coli*?

(1) To denature the DNA so it becomes single-stranded.

(2) To kill the bacteria that have not taken up the plasmid.

(3) To transiently change the membrane permeability to allow rDNA entry.

(4) To activate the 'ori' site for immediate replication.

10) Consider the following steps in the isolation of DNA: A. Treatment with Protease B. Cell wall digestion by enzymes C. Precipitation by chilled ethanol D. Removal of RNA by Ribonuclease What is the correct chronological order?

(1) B → A → D → C

(2) B → D → A → C

(3) A → B → C → D

(4) D → A → B → C

11) Which of the following statements is INCORRECT regarding the methods of introducing alien DNA into host cells?

(1) In micro-injection, DNA is directly injected into the nucleus of animal cells.

(2) Biolistics is suitable for plants using high-velocity micro-particles.

(3) 'Disarmed' pathogens are allowed to infect the cell to transfer rDNA.

(4) Competent host cells are prepared by treating them with monovalent cations like Sodium.

12) A researcher successfully bombards plant cells with gold micro-particles coated with DNA. However, the DNA fails to replicate in the daughter cells. What is the most likely reason for this failure?

(1) The DNA was not coated with Tungsten.

(2) The DNA fragment lacked an 'origin of replication' recognized by the host.

(3) The cells were not treated with chilled ethanol before bombardment.

(4) The researcher used micro-injection instead of biolistics.

13) In recombinant DNA technology, the passenger DNA fragment and cloning vector are usually cut with the same restriction enzyme so that:

(1) The vector becomes larger

(2) Compatible sticky ends are produced for efficient ligation

(3) The host cell replicates faster

(4) The passenger DNA becomes circular

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14) Which of the following would be the most likely outcome if a scientist forgets to add Ethidium Bromide during the gel electrophoresis process?

(1) The DNA fragments will not separate.
(2) The DNA fragments will move in the opposite direction.
(3) The DNA bands will not be visible under UV light.
(4) The DNA will be degraded by the UV light.

15) A recombinant plasmid containing passenger DNA fails to replicate in the host cell. Which feature of the cloning vector is most likely missing or damaged?

(1) Restriction site
(2) Ori (origin of replication)
(3) Passenger DNA
(4) Antibiotic resistance gene

16) "Disarmed" vectors are used in biotechnology to:

(1) Deactivate the host's immune system.
(2) Ensure the vector can no longer replicate in the host.
(3) Prevent the vector from causing disease while retaining its ability to transfer DNA.
(4) Strip the DNA of its phosphate backbone to make it less acidic.

17) Which of the following is common to both Micro-injection and Biolistics?

(1) Both are used specifically for bacterial transformation.
(2) Both are methods of "vector-less" or direct gene transfer.
(3) Both require the use of gold or tungsten particles.
(4) Both require the host to be treated with $CaCl_2$.

18) Identify the correct sequence of treatments for introducing recombinant DNA into a bacterial host:

(1) Ice → Divalent cation treatment → Heat shock (42°C) → Ice
(2) Divalent cation treatment → Ice → Heat shock (42°C) → Ice
(3) Heat shock (42°C) → Divalent cation treatment → Ice → Ice
(4) Divalent cation treatment → Heat shock (42°C) → Ice → 37°C incubation

19) Consider the following statements regarding the isolation of genetic material:
(A) Ribonuclease is used to remove RNA.
(B) Chilled ethanol is added to precipitate the purified DNA.
(C) DNA is removed as fine threads by a process called spooling.
(D) Proteases are used to remove histone proteins associated with DNA.

Which of the following combinations of statements are correct according to NCERT?

(1) A and B only

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(2) A, B, and C only

(3) B, C, and D only

(4) A, B, C, and D

20) How many phosphodiester bonds are broken by a single EcoRI molecule to produce two fragments from one linear DNA duplex containing one recognition site?

(1) 1

(2) 2

(3) 4

(4) 6

21) In the diagram showing the formation of recombinant DNA, which enzyme is responsible for joining the "sticky ends" via sugar-phosphate backbones?

(1) Restriction Endonuclease

(2) DNA Polymerase

(3) DNA Ligase

(4) Alkaline Phosphatase

22) During DNA isolation, why are molecules like "Proteins" and "Polysaccharides" treated with specific enzymes?

(1) To ensure DNA remains in a double-helical state.

(2) Because DNA is intermixed with these macromolecules in the cell.

(3) To catalyze the action of restriction enzymes later.

(4) To prevent the DNA from precipitating.

23) If a restriction enzyme makes a cut between 'G' and 'A' in the sequence 5' - GAATTC - 3', what will be the resulting overhanging sequence on that specific strand?

(1) CTTAA

(2) AATTC

(3) GAATT

(4) TTAAG

24) Assertion (A): EcoRI cuts the DNA between bases G and A only when the sequence GAATTC is present in the DNA.
Reason (R): Restriction endonucleases inspect the length of a DNA sequence to find their specific recognition sequence.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

25) When cutting DNA with a restriction enzyme, how many phosphodiester bonds are broken in one

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double-stranded DNA molecule to produce sticky ends?

(1) One
(2) Two
(3) Four
(4) Six

26) A researcher wants to transform a plant cell. Which of the following combinations of tools/methods would be most appropriate?

(1) *EcoRI* and Retrovirus
(2) Ti plasmid and Biolistics
(3) Lysozyme and Heat Shock
(4) pBR322 and Micro-injection

27) Which of the following would be the most likely outcome if a scientist forgets to add Ethidium Bromide during the gel electrophoresis process?

(1) The DNA fragments will not separate.
(2) The DNA fragments will move in the opposite direction.
(3) The DNA bands will not be visible under UV light.
(4) The DNA will be degraded by the UV light.

28) In the screening of recombinants using the *lacZ* gene, what is the appearance of the non-recombinant colonies on a medium containing X-gal?

(1) Colorless
(2) Blue
(3) Red
(4) Orange

29) Which of the following is not correct about pBR 322 and such vectors?

(1) It was constructed by using DNA derived from naturally occurring plasmids of E coli.
(2) It has two drug resistance genes-tet^R and amp^R.
(3) It was developed by Bolivar and Rodriguez.
(4) Selectable markers present in it can differentiate recombinants from non-recombinants on the basis of their inability to produce colour in the presence antibiotic.

30) Identify the correct set of statements :
(a) When different DNAs are cut by same restriction enzyme, the resultant DNA fragments have the same kind of sticky ends.
(b) The separated bands of DNA are cut out from the agarose gel and extracted from the gel piece. This step is known as elution.
(c) The RNA can be removed by treatment with protease.
(d) If any protein encoding gene is expressed in a heterologous host it is called a recombinant protein.

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Choose the correct answer from options given below :

(1) a and b only

(2) b, c and d only

(3) a, b and d only

(4) a and d only

31) Identify the correct set of statements :
(a) In gel electrophoresis, the most commonly used matrix is agarose.
(b) Plasmids and bacteriophages do not have the ability to replicate within bacterial cells.
(c) Transformation is a procedure through which a piece of DNA is introduced in a host bacterium.
Choose the correct answer from options given below :

(1) a and b only

(2) a and c only

(3) b and c only

(4) a, b and c

32) Given below are four statements pertaining to separation of DNA fragments using gel electrophoresis. Identify the incorrect statements.
(a) DNA is negatively charged molecule and so it is loaded on gel towards the Anode terminal
(b) DNA fragments travel along the surface of the gel whose concentration does not affect movement of DNA.
(c) Smaller the size of DNA fragment larger is the distance it travels through it.
(d) Pure DNA can be visualized directly by exposing UV radiation.
Choose correct answer from the options given below

(1) (a), (c) and (d)

(2) (a), (b) and (c)

(3) (b), (c) and (d)

(4) (a), (b) and (d)

33) Which one of the following base sequences in DNA can be easily cut by *Eco*RI ?

(1) 5' _____ CACGTA _____ 3'
3' _____ CTCAGT _____ 5'

(2) 5' _____ CGTTCG _____ 3'
3' _____ ATGGTA _____ 5'

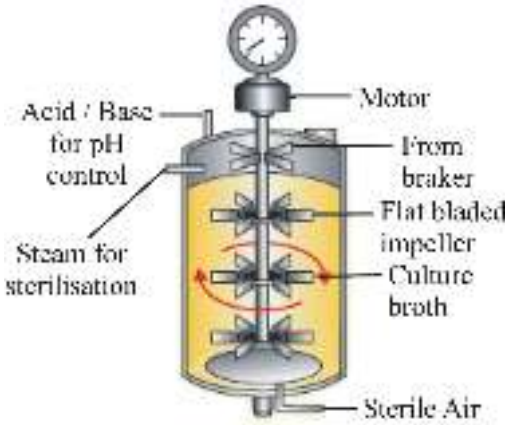
(3) 5' _____ GATATG _____ 3'
3' _____ CTACTA _____ 5'

(4) 5' _____ GAATTC _____ 3'
3' _____ CTTAAG _____ 5'

34) Given below in the figure of one of the most commonly used bioreactor.

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In this flat bladed impeller helps in :-

(1) Maintaining the pH of medium

(2) Stirring the liquid medium

(3) Regulating temperature of medium

(4) Both (1) & (3)

35) After the restriction digestion of a genomic DNA, the fragments are separated according to their size through.....provided by agarose gel during electrophoresis. (Fill in the gap with following options)

(1) PCR

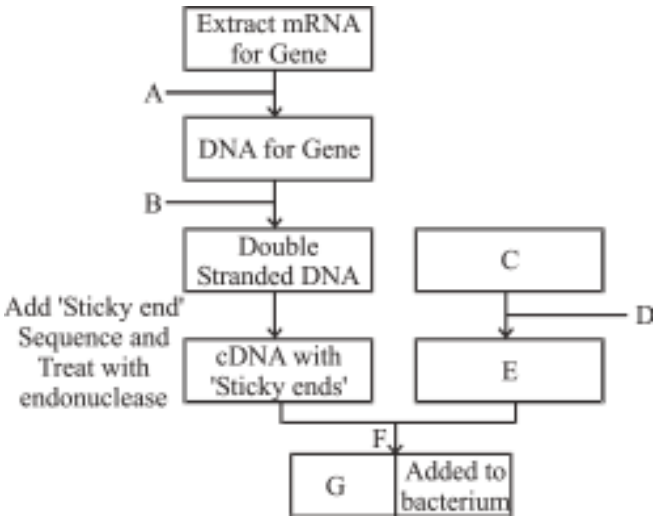
(2) Spooling method

(3) Sieving effect

(4) DNA precipitation

36)

Identify the labelled items A, B, C, D, E, F and G in the diagram below from the list I to VII :-



(I) DNA polymerase

(II) Plasmid

(III) Plasmid with 'sticky ends'

(IV) DNA ligase

(V) Restriction endonuclease

(VI) Recombinant DNA

(VII) Reverse transcriptase The correct components are

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	A	B	C	D	E	F	G
(1)	VII	I	II	V	III	IV	VI
(2)	VII	VI	V	IV	III	II	I
(3)	VII	V	III	I	II	IV	VI
(4)	I	II	IV	VI	III	V	VII

(1) 1

(2) 2

(3) 3

(4) 4

37) Arrange the steps of rDNA technology in correct order:-
I. Extraction of the desired gene product.
II. Amplification of gene of interest.
III. Isolation of desired DNA fragment.
IV. Ligation of DNA fragment into vector.
V. Insertion of rDNA into host.
Correct order is :-

(1) I, II, III, IV, V

(2) V, IV, III, II, I

(3) III, II, IV, V, I

(4) III, IV, II, I, V

38) Match the column:

	Column-I		**Column-II**
(i)	Restriction enzyms	(p)	Eco RI
(ii)	Restriction site	(q)	Selectable marker
(iii)	Plasmid	(r)	Important vectors
(iv)	Ampicillin	(s)	Works on palindrome

(1) (i-q), (ii-s), (iii-r), (iv-p)

(2) (i-s), (ii-p), (iii-r), (iv-q)

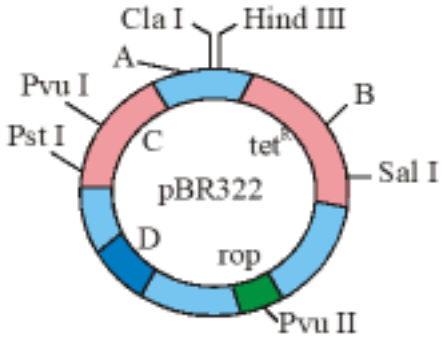
(3) (i-r), (ii-q), (iii-p), (iv-s)

(4) (i-p), (ii-s), (iii-r), (iv-q)

39) Identify A, B, C and D in the given figure of *E.coil* cloning vector pBR322 and select the correct option :-

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	A	B	C	D
(1)	Hind I	Eco R I	amp^R	ori
(2)	Hind I	Bam H I	kan^R	amp^R
(3)	Bam H I	Pst I	ori	amp^R
(4)	Eco R I	Bam H I	amp^R	ori

(1) 1

(2) 2

(3) 3

(4) 4

40) Which of the following are the steps of downstreaming.
(a) Quality control
(b) Clinical trial
(c) Purification
(d) Maintenance of optimal growth conditions
(e) Separation
(f) Production of recombinant protein

(1) a, b, c, d, f

(2) a, b, c, e, f

(3) a, b, c, e

(4) a, b, c, d, e

41) The key tools required for the recombinant DNA technology are
I. restriction enzymes.
II. Polymerase enzymes.
III. Host organism ligases.
IV. vectors.
V. Host organisms, select the correct option.

(1) I, II and III only

(2) I, III, IV and V only

(3) I, II, III and V only

(4) I, II, III, IV and V

42) In bioprocess engineering, stirred-tank bioreactors provide better aeration than shake flasks.

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If a culture of recombinant E. coli expressing a oxygen-sensitive protein is grown in a sparged stirred tank bioreactor with excessive agitation, what might be the unintended consequence on protein yield?

(1) Increased biomass but denatured protein due to shear stress.

(2) Reduced contamination but lower copy number.

(3) Better pH control but plasmid loss.

(4) Higher temperature stability but no effect on yield.

43) Which of the following statements is/are incorrect.
(a) The digested DNA fragments can be separated by a technique known as gel electrophoresis.
(b) The separated bands of DNA are cut out from the agarose gel and extracted from the gel is known as spooling.
(c) The DNA fragments purified are used in constructing recombinant DNA by joining them with cloning vectors.
(d) The most commonly used matrix is agarose gel which is an artificial polymer extracted from sea weeds.

(1) a and b

(2) b and d

(3) b and c

(4) c and d

44) Which of the following is true regarding insertional inactivation of β–galactosidase coding sequence?

(1) The recombinant colonies give blue colour in presence of chromogenic substrate.

(2) The non-recombinant colonies do not produce any colour.

(3) The recombinant colonies do not produce any colour even in the presence of chromogenic substrate.

(4) The non-recombinant colonies give blue colour even in the absence of chromogenic substrate

45) Assertion A: For large-scale production of a recombinant protein, the cells harboring the cloned gene of interest need to be cultured in a continuous system.
Reason R: A continuous culture system allows cells to remain in their most active log/exponential phase by adding fresh medium and draining used medium.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

46) Read the following statements :
(i) Wheat, maize barley, castor and brinjal seeds are endospermic or albuminous
(ii) Most zygotes divide only after certain amount of endosperm is formed
(iii) Though the seeds differ greatly, the early stages of embryo development is similar in both monocots and dicots
(iv) The zyogte divides to form proembryo and subsequently globular, heart-shape and mature

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embryo
(v) Epicotyl and hypocotyl are absent in monocot seeds
Which of the above statements are correct ?

(1) (i), (ii), (iii) and (iv)
(2) (i), (ii) and (iii)
(3) (i), (iii) and (v)
(4) (iii), (iv) and (v)

47) How many of the following are incorrect
(A) Flowers are morphological and embryological marvels and the sites of sexual reproduction.
(B) Much before the actual flower is seen on a plant, the decision that the plant is going to flower has taken place.
(C) Flowers have always been used as symbols for conveying important human feelings.
(D) Human beings have had a intimate relationship with flowers since time immemorial.
(E) Several hormonal and structural changes are initiated which lead to the differentiation and further development of floral primordium.
(F) Flowers are objects of aesthetic, ornamental, social, religious and cultural values.

(1) 0
(2) 1
(3) 2
(4) 3

48) The middle layer of the anther wall
a. Nourishes developing pollen grains
b. Helps in the dehiscence of anther due hygroscopic nature
c. Has water resistant fibrous bands
d. Degenerates at maturity
e. Has cells with more than two complete sets chromosomes
The correct one(s) is/are

(1) All are correct except b
(2) a, b and d
(3) c, d and e
(4) d only

49) The disadvantage of cleistogamy is that

(1) Offspring produced have limited genetic diversity
(2) It produces limited number of pollen grains
(3) There is a little probability of flower to get pollinated
(4) There is always a need of pollinator

50) Read the following statements and mark them as True (T) or False (F).
a. Coconut water from tender coconut has thousands of free nuclei.
b. Both endosperm and perisperm are similar in ploidy level and their origin.
c. Coleoptile is a hollow foliar structure that encloses leaf primordia in embryo.

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d. Single, large shield shaped cotyledon of wheat embryo is known as epiblast.

	a	**b**	**c**	**d**
(1)	T	F	F	F
(2)	T	F	T	F
(3)	T	F	F	T
(4)	F	T	F	F

(1) 1

(2) 2

(3) 3

(4) 4

51) A flower produced pollen grains that lacked exine completely. Which property would be lost immediately?

(1) Ability to undergo meiosis

(2) Resistance against environmental stress

(3) Formation of vegetative nucleus

(4) Ability to carry male gametes

52)

Plants with one or few ovules are generally pollinated by–

(1) Wind

(2) Bees

(3) Birds

(4) Butterflies

53) *Amorphophallus* flower provide floral rewards to the pollinator in the form of :-

(1) Beautiful colour

(2) Providing safe place to lay eggs

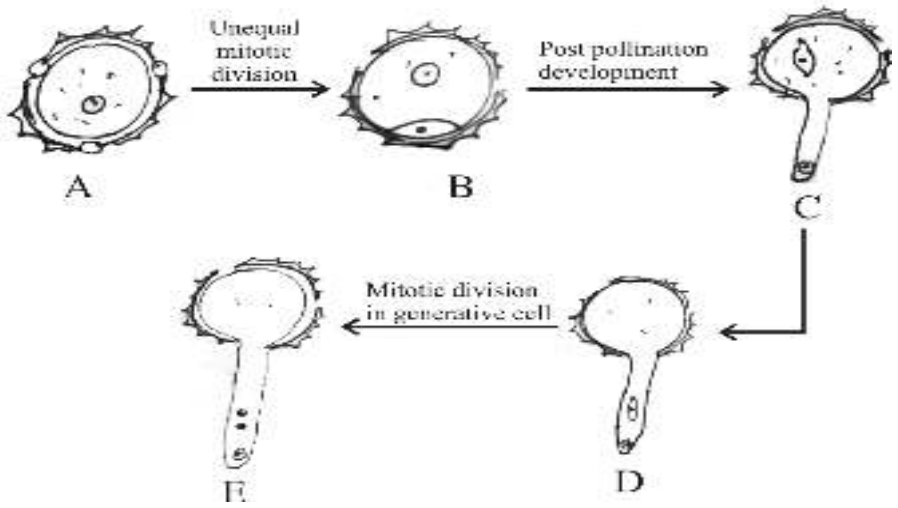
(3) Nectar

(4) Providing safe place for asexual reproduction

54)

The figure given below represents the various steps of microgametogenesis. Select the option giving correct identification together with what it represents?

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(1) (C) represents mature male gametophyte
(2) (A) Represents mature pollen grain
(3) (E) Represents mature male gametophyte
(4) (B) Represents mature male gametophyte

55) Genetic Map is one that :

(1) Establishes sites of the genes on a chromosome
(2) Establishes the various stages in gene evolution
(3) Shows the stages during the cell division
(4) Shows the distribution of various species in a region

56) One of the genes present exclusively on the X-chromosome in humans is concerned with

(1) Baldness
(2) Red green colour blindness.
(3) Facial hair/Moustaches in males.
(4) Sickle cell anemia

57) A normal woman has a haemophilic son besides three perfectly normal daughters. The genotypes of the woman and her husband with respect to haemophilia would respectively be:-

(1) $X^{+}X^{+}$ and $X^{h}y$
(2) $X^{h}X^{+}$ and $X^{h}y$
(3) $X^{h}X^{h}$ and $X^{+}y$
(4) $X^{h}X^{+}$ and $X^{+}y$

58) A molecular biologist analysed a nucleic acid sample and found that it contained the N_2 bases in following proportion A = 20%, T = 15%, C = 35%, G = 30% This data indicates that

(1) Nucleic acid in the sample is single stranded DNA
(2) Nucleic acid in the sample is double stranded DNA
(3) Nucleic acid in the sample is double stranded RNA

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(4) Nucleic acid in the sample is single stranded RNA

59) Cell having YYRr genes produce 1000 gametes how many of them might have both dominant factors ?

(1) 1000

(2) 500

(3) 250

(4) None of the above

60) The test cross of an F1 individual with genotype (++/ab) produced the following offsprings :
++/ab 10
ab/ab 10
+a/ab 40
+b/ab 40
Based on this data predict the configuration of parents :

(1) Cis-configuration

(2) Trans-configuration

(3) Both the above

(4) Complete Linkage

61) Choose the correct option :-

(a)	Linkage theory	(i)	A.Sturtevant
(b)	Chromosomal theory of inheritance	(ii)	Henking
(c)	X-chromosome discovery	(iii)	Sutton and Boveri
(d)	Linkage map	(iv)	T.H.Morgan

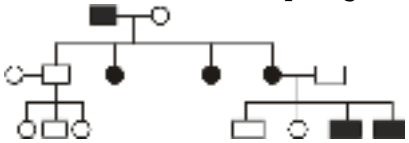
(1) a – (i), b – (ii), c – (iii), d – (iv)

(2) a – (iv), b – (iii), c – (ii), d – (i)

(3) a – (iv), b – (iii), c – (i), d – (ii)

(4) a – (iv), b – (i), c – (ii), d – (iii)

62) Given below is a pedigree of a family suffering from a genetical disorder.



After observing the inheritance pattern find out that which of the following disorder can produce such type of pedigree :-

(1) Hypertrichosis

(2) Colour blindness

(3) Haemophilia

(4) Pseudoricketes

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63) Match the following :-

A	Down's syndrome	a	Terminal deletion of short arm of chromosome-5
B	Philadelphia syndrome	b	Autosomal aneuploidy
C	Cat cry syndrome	c	Chronic myeloid leukemia

(1) A-b, B-c, C-a

(2) A-b, B-a, C-c

(3) A-a, B-c, C-b

(4) A-a, B-b, C-c

64) Assertion : Lac operon in the prototype operon in the bacteria.
Reason : The operon is regulated by the amount of lactose in the medium where bacteria is grown.

(1) Both Assertion(A) and reason(R) are true and reason(R) is the correct explanation of the assertion(A)

(2) Both Assertion(A) and reason(R)are true, but reason(R)is not the correct explanation of the Assertion(A)

(3) Assertion(A) is true, but reason(R) is false

(4) Assertion(A) is false but Reason(R) is true

65) Statement I: Regulation of gene expression is controlled only by metabolic or physiological condition.
Statement II: The actual structure of tRNA was explained by using cover leaf model.

(1) Both statements are correct

(2) Both the statements are incorrect

(3) Only statement I is correct

(4) Only statement II is correct

66) Statement I: Regulatory sequence can authentically be called as regulatory genes.
Statement II: Regulatory sequences code for RNA and proteins.

(1) Both statements are correct

(2) Both statements are incorrect

(3) Only statement I is correct

(4) Only statement II is correct

67) Arrange the following in the order of their discovery from older to recent
(i) Discovery of Down's syndrome
(ii) Discovers of X body
(iii) Nuclein by meischer
(iv) postulating genetic code
(v) existence of tRNA
(vi) chromosome 1 sequencing

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(1) (i), (ii), (iii), (iv), (v) and (vi)
(2) (ii), (iii), (i), (iv), (v) and (vi)
(3) (ii), (iii), (i), (v), (iv) and (vi)
(4) (ii), (i), (iii), (v), (iv) and (vi)

68) Match the Following

Column A – Parental Genotypes	**Column B – Possible Gametes**
A. RrYy	a. RY, Ry, rY, ry
B. RRYY	b. RY
C. rryy	c. ry
D. RRYy	d. RY, Ry

Select the correct match:
(1) A–a, B–b, C–c, D–d
(2) A–b, B–a, C–d, D–c
(3) A–d, B–c, C–b, D–a
(4) A–a, B–a, C–c, D–a

69) In double helix of DNA, the two strands are

(1) Coiled around a common axis
(2) Coiled differently
(3) Coiled over protein sheath
(4) Coiled around each other

70) Wilkins & Franklin studied DNA by –

(1) Autoradiography
(2) Chromatography
(3) X-Ray Crystallography
(4) None

71) A DNA molecule in which both strands have radioactive thymidine is allowed to duplicate in an environment containing non-radioactive thymidine. What will be the exact number of DNA molecules that contain some radioactive thymidine after 3 duplication ?

(1) One
(2) Two
(3) Four
(4) Eight

72) Some of the steps of DNA fingerprinting are given below. Identify their correct sequence from the options given.

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A. Electrophoresis of DNA fragments
B. Hybridisation with DNA probe
C. Digestion of DNA by restriction endonucleases
D. Autoradiography
E. Blotting of DNA fragments to nitrocellulose membrane

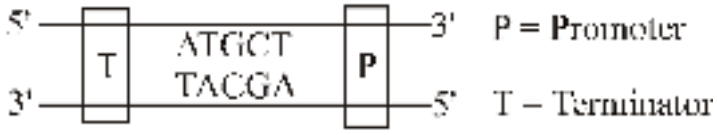
(1) C – A – B – E – D (2) (3) (4)

(2) C – A – E – B – D

(3) A – E – C – B – D

(4) A – C – E – D – B

73)



The m-RNA produced by transcribtion of above given DNA is :-

(1) 5' AUGCU 3'

(2) 5' AGCAU 3'

(3) 3' UCGUA 5'

(4) 5' UACGA 3'

74) If the sequence of coding strand of DNA is 5`GGCCTTAATGGCG3`, what will be the sequence of mRNA formed from its template strand?

(1) 5`CCGGAAUUACCGC3`

(2) 3`GGGGAATTACGCC5’

(3) 3’GGCCAAUUACCGC5`

(4) 5’GGCCUUAAUGGCG3`

75) Which statement is incorrect for VNTR sequence ?
(i) VNTR belongs to a class of satellite DNA referred to as micro satellite
(ii) Copy number of VNTR varies from chromosome to chromosome in an individual
(iii) The size of VNTR varies in size from 0.1 to 20 Kb
(iv) The number of repeat of VNTR show very low degree of polymorphism

(1) (i) and (ii)

(2) (ii) and (iii)

(3) (i), (ii), (iii) and (iv)

(4) (i) and (iv)

76) DNA replication in *E. coli* takes place during

(1) G_1 phase

(2) S phase

(3) G_0 phase

(4) No such specific phase

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77) Prior to the work of Oswald Avery, Colin MacLeod, and Maclyn McCarty (1933-44), the genetic material was thought to be-

(1) Cellulose
(2) RNA
(3) Protein
(4) Lipid

78)

During translation, peptide bond is formed between:-

(1) NH_2 - group of P-site amino acid and –COOH group of A - site amino acid
(2) –COOH group of P-site amino acid and –NH_2 group of A – site amino acid
(3) – NH_2 group of P–site amino acid and –NH_2 group of A–site amino acid
(4) –COOH group of P–site amino acid and –COOH group of A–site amino acid

79) How many of the following is/are correct statement ?
(A) In capping an unusual nucleotides is added to the 3' end of hn RNA
(B) In tailing adenylated residues are added at 3' end in a template independent manner.
(C) The untranslated regions (UTR) are required for efficient DNA replication process.
(D) The DNA-dependent DNA polymerases catalyse polymerisation only in one direction that is 5' → 3'.

(1) Four
(2) Three
(3) Two
(4) One

80) The okazaki fragments are joined together by an enzyme?

(1) DNA ligase
(2) DNA polymerase
(3) Topoisomerase
(4) Helicase

81) If normal E.Coli is allow to grow for 80 minutes in $^{15}NH_4Cl$ medium then what would be the proportion of hybrid and heavy density DNA molecules

(1) 1 : 7
(2) 7 : 1
(3) 14 : 2
(4) 1 : 4

82) Which property of genetic code ensures that **mutation in the third base of codon often does not change the amino acid**?

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(1) Universality
(2) Degeneracy
(3) Unambiguity
(4) Non-overlapping nature

83) Consider a eukaryotic cell synthesizing a polypeptide chain 150 amino acids long. The mature mRNA sequence originally had an intron of 90 nucleotides. If an error during splicing leaves this intron intact, and assuming no premature stop codons are introduced, what will be the length of the resulting polypeptide synthesized, and which ribosomal unit translates it?

(1) 180 amino acids; 80S ribosomes in the cytoplasm
(2) 150 amino acids; 70S ribosomes in the nucleoplasm
(3) 180 amino acids; 70S ribosomes in the mitochondria
(4) 240 amino acids; 80S ribosomes in the cytoplasm

84) It is found that the genetic code in the bacteriophage Φ×174 is overlapping. The total number of N-bases required to synthesise a polypeptide chain of 2001 amino acids by the genetic material of Φ×174 in the appropriate host cell will be:

(1) 667
(2) 2003
(3) 6003
(4) 6006

85) During elongation step of translation, the total no. of GTPs required to synthesise a polypeptide of 109 amino acids is:

(1) 218
(2) 217
(3) 216
(4) 215

86) New high-throughput technologies (such as RNA-sequencing and DNA microarrays) are required for:

(1) To study many genes at a time on a broader scale.
(2) To eliminate the need for reverse transcription during RNA profiling.
(3) To increase the use of conventional molecular approaches such as Sanger's sequencing.
(4) To limit the genomic investigations for repeated sequences only.

87) Polysomes are —
(i) A single mRNA is translated simultaneously by several ribosomes.
(ii) Many copies of same polypeptide are required.
(iii) Many different types of polypeptides are required from the same mRNA.
(iv) Found in Prokaryotes only.
(v) Found in eukaryotes only.
(vi) Found in both prokaryotes and eukaryotes.

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How many statements are correct?

(1) Two
(2) Three
(3) Four
(4) Five

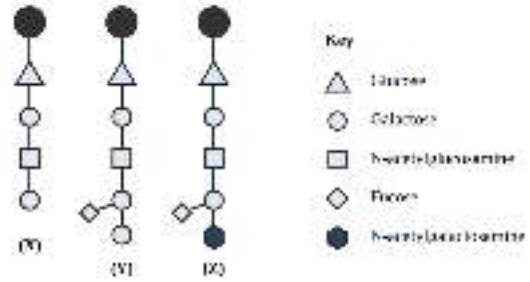
88) Glucose is preferred source of energy for E. coli over lactose. A medium is supplemented with both glucose and lactose to grow a mutant strain of E. coli in which the y-gene is mutated, resulting in the formation of a non-functional enzyme. The possible outcome on the functioning of the lac operon will be:

(1) In the beginning operon will be in OFF state then ON.
(2) The operon will not be functional at all.
(3) The operon will be activated during glucose metabolism then become non-functional.
(4) The E. coli will not face any scarcity of carbon source as glucose is present.

89) Read the following statements regarding the sequencing methodology in the Human Genome Project (HGP):
(i) Automated DNA sequencers functioned on the principle of a method developed by Frederick Sanger.
(ii) The generated DNA sequences were aligned and assembled on the basis of overlapping regions present in them.
(iii) Alignment of these sequences was humanly not possible and was carried out using specialized computer-based programs.
(iv) Frederick Sanger designed the computer software programs used for aligning the sequenced fragments.
Which of the statements given above are correct?

(1) (i) and (ii) only
(2) (i), (ii) and (iii) only
(3) (ii) and (iv) only
(4) (i), (ii), (iii) and (iv)

90) Given below is the schematic representation of sugar moieties of blood group ABO antigens :



Select the correct option that represents blood group A antigen :

(1) X
(2) Y
(3) Z
(4) None of these

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ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	3	3	3	2	4	2	3	1	2	2	1	4	4	2	3	2	3	1	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	3	4	4	4	2	4	2	4	2	1	1	1	2	2	3	2	4	3	2
Q.	41	42	43	44	45															
A.	2	1	3	3	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	2	2	4	2	3	2	4	3	3	2	3	2	1	2	4	2	4	2	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	1	4	3	1	2	3	2	1	4	1	4	2	2	4	2	4	4	2	3
Q.	86	87	88	89	90															
A.	3	3	2	3	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	2	2	1	2	4	3	4	3	2	4	2	2	3	2	3	2	2	4	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	2	2	1	2	2	3	2	4	3	2	4	4	2	3	1	3	4	4	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	1	2	3	1	1	1	4	1	2	2	1	2	3	1	3	4	1	2	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	4	1	3	2	2	4	1	1	3	2	2	2	4	4	4	3	2	3	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	2	1	2	3	1	2	2	2	3										

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SOLUTIONS

PHYSICS

1) $W_2 \rightarrow 1 = q(V_1 - v_2)$

$$\left[\left(\frac{KQ_1}{R} + \frac{KQ_2}{\sqrt{R^2+R^2}}\right) - \left(\frac{KQ_2}{R} + \frac{KQ_1}{\sqrt{R^2+R^2}}\right)\right]$$

$$\frac{Kq(Q_1 - Q_2)}{R}\left(1 - \frac{1}{\sqrt{2}}\right)$$

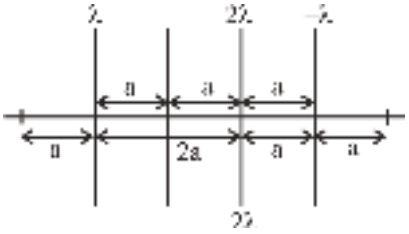
2) $E = \frac{\sigma}{2\epsilon_0}\left(1 - \frac{x}{\sqrt{R^2+x^2}}\right)$

$x = 0,\ E_1 = \frac{\sigma}{2\epsilon_0}$

$x = ,\ E_2 = \frac{\sigma}{2\epsilon_0}\left(1 - \frac{x}{\sqrt{R^2+x^2}}\right)$

$\frac{E_2 - E_1}{E_1} \times 100 = -86.6\%$

3)



at x = –2a

$$-\frac{2k\lambda}{a}\hat{i} - \frac{2k \times 2\lambda}{3a}\hat{i} + \frac{2k\lambda}{4a}\hat{i}$$

$$2k\lambda\left[\frac{-12-16+3}{12a}\right]\hat{i}$$

$$-\frac{25k\lambda}{6a}\hat{i}$$

At x = 0

$$\frac{2k\lambda}{a}\hat{i} - \frac{2k \times 2\lambda}{a}\hat{i} + \frac{2k\lambda}{3a}\hat{i}$$

$$\frac{3k\lambda}{a}\hat{i} - \frac{4k}{a}\hat{i} = -\frac{k\lambda}{a}\hat{i}$$

At x = $\frac{3a}{2}$

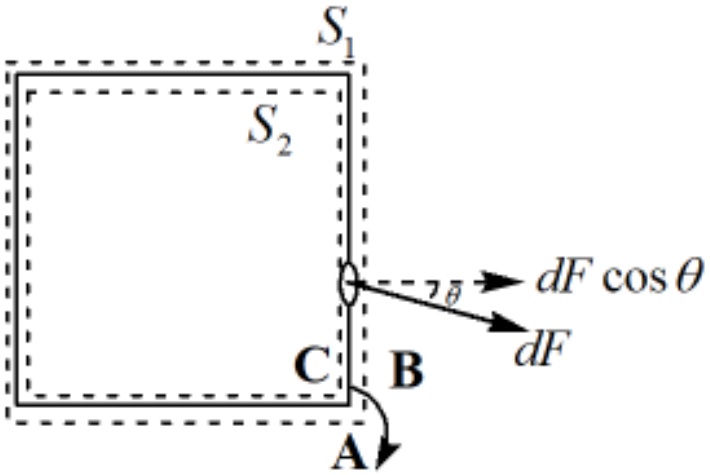
$$\frac{4k\lambda}{5a}\hat{i} + \frac{2k \times 2\lambda}{\frac{a}{2}}\hat{i} + \frac{k\lambda}{a}\hat{i}$$

$$\frac{4 + 40 + 20}{5a}$$

4) Consider two closed Gaussian surfaces S_1 just enclosing the tube and other S_2 just inside the

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cube. Consider one of the faces of the cube (say A)



Let ϕ be the flux through face B of cubical Gaussian surface S_1 due to remaining five faces of cube and ϕ' be the flux through the same face B due to face A of the cube.

Then $\phi + \phi' = \frac{1}{6}\left(\frac{6Q}{\epsilon_0}\right) = \frac{Q}{\epsilon_0}$

Flux through C of cubical Gaussian surface S_2 due to face A of the cube is also ϕ' and flux due to remaining five faces is ϕ in opposite directions

Net flux through face C of cubical Gaussian surface S_2 is zero ($\because$ charge enclosed by S_2 is zero)

B and C are faces of cubical Gaussian surfaces S_1 and S_2 on either side of A respectively.

$$\phi - \phi' = 0 \Rightarrow \phi = \phi' \Rightarrow 2\phi = \frac{Q}{\epsilon_0} \Rightarrow \phi = \frac{Q}{2\epsilon_0}$$

Consider a small element of area dA of face A. E be the intensity of electric field due to remaining faces at this element

Force on the element is $dF = E\,dq = E\sigma dA$

$\therefore$ Force on the face A is $F = \int dF\cos\theta = \int E\sigma dA\cos\theta = \sigma\int d\phi = \sigma\phi$ where ϕ flux through face A due to remaining five faces

$\therefore$ $F = \frac{Q}{A}\times\frac{Q}{2\epsilon_0} = \frac{Q}{d^2}\times\frac{Q}{2\epsilon_0} = \frac{Q^2}{2\epsilon_0 d^2}$

5) Potential difference between two points at perpendicular distances r_1, r_2 from a charged wire is $V_2 - V_1 = \frac{\lambda}{2\pi\varepsilon_0}\ln\frac{r_1}{r_2}$.

Where r_1, r_2 are perpendicular distances of initial and final points from the wire.

$W = (V_2 - V_1)Q$

$r_1 = \sqrt{2}\,m,\ r_2 = \sqrt{2}\,m$ for wire along X- axis, $r_1 = \sqrt{2}\,m,\ r_2 = 1\,m$ for both the wires along Y& Z- axes

$$\therefore W = \sum\frac{\lambda}{2\pi\varepsilon_0}\ln\frac{r_1}{r_2} = \frac{2\lambda}{2\pi\varepsilon_0}\ln\frac{\sqrt{2}}{\sqrt{2}} + \frac{\lambda}{2\pi\varepsilon_0}\ln\frac{\sqrt{2}}{1} + \frac{3\lambda}{2\pi\varepsilon_0}\ln\frac{\sqrt{2}}{1}$$

$$= \frac{4\lambda}{2\pi\varepsilon_0}\ln\sqrt{2} = \frac{\lambda}{\pi\varepsilon_0}\ln 2$$

6) Charge on $a_1 = (r_1\theta)\lambda$ and Charge on $a_2 = (r_2\theta)\lambda$ $\therefore$ ratio of charges $= \frac{r_1}{r_2}$

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NOVA @examlix

At point A intensity of electric field produced by a_1 is $E_1 = \frac{K(r_1\theta)\lambda}{r_1^2} = \frac{K\theta\lambda}{r_1}$

At point A intensity of electric field produced by a_2 is $E_2 = \frac{K(r_2\theta)\lambda}{r_2^2} = \frac{K\theta\lambda}{r_2}$

As $r_2 > r_1 . E_1 > E_2$ and net field at A is towards a_2

At point A potential due to a_1 is $V_1 = \frac{K(r_1\theta)\lambda}{r_1} = K\theta\lambda$

At point A potential due to a_2 is $V_2 = \frac{K(r_2\theta)\lambda}{r_2} = K\theta\lambda \quad \therefore V_1 = V_2$

7) $U = \frac{qQ}{a+x} + \frac{qQ}{a-x} = \frac{2qQ}{a}\left(1 - \frac{x^2}{a^2}\right)^{-1}$

$\approx \frac{2qQ}{a}\left(1 + \frac{x^2}{a^2}\right) \quad (\because x << a)$

$\Delta u = U - U_0 = \frac{2qQ}{a}\left(1 + \frac{x^2}{a^2}\right) - \frac{2qQ}{a}$

$= \frac{2qQ}{a^3}\left(x^2\right)$

$\Delta U \propto x^2$

8)

Question Explanation:
The question asks the electric field at 5 cm from the center of a uniformly charged sphere when the electric field at 20 cm from the center is E.

Concept:
Electric Field of a Uniformly Charged Sphere

Outside the sphere (r > R), $E \propto \frac{1}{r^2}$.
Inside the sphere (r < R), $E \propto r$.

Solution:
Given radius of sphere R = 10 cm.
At r = 20 cm (outside the sphere):

$E_{20} = \frac{kQ}{(20)^2} = E$

At r = 5 cm (inside the sphere), electric field is

$E_{inside} = \frac{kQr}{R^3}$

Thus,

$E_5 = \frac{kQ(5)}{10^3}$

Now compare with E_{20}:

$E_{20} = \frac{kQ}{20^2} = \frac{kQ}{400}$

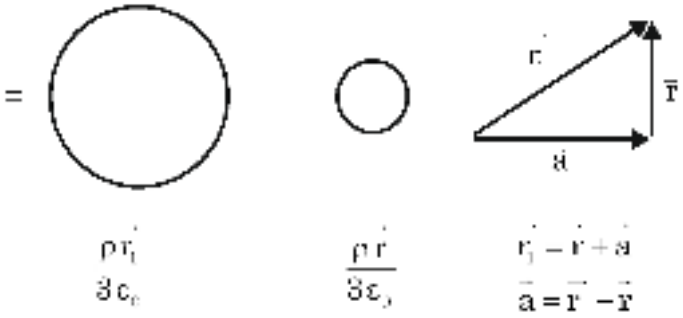
$\frac{E_5}{E_{20}} = \frac{(5/1000)}{(1/400)} = 2$

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So,
$E_5 = 2E$

Final Answer:
2E

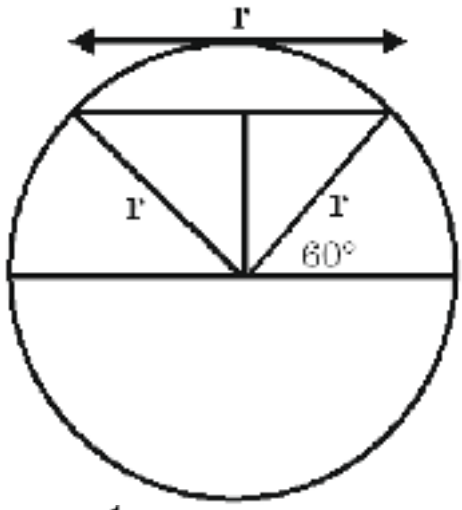
9) Electric fued



$\frac{\rho \vec{r_1}}{3\varepsilon_0}$ $\frac{\rho \vec{r}}{3\varepsilon_0}$ $\vec{r_1} - \vec{r} + \vec{a}$

$\vec{a} = \vec{r} - \vec{r}$

$$E_{net} = \frac{\rho \vec{r_1}}{3\varepsilon_0} - \frac{\rho \vec{r}}{3\varepsilon_0}$$
$$= \frac{\rho}{3\varepsilon_0}(\vec{r_1} - \vec{r})$$
$$= \frac{\rho}{3\varepsilon_0}\vec{a}$$



$$s = ut + \frac{1}{2}at^2$$
$$t = \sqrt{\frac{2s}{a}} = \sqrt{\frac{2r}{qE/m}}$$
$$= \sqrt{\frac{2rm.3\varepsilon_0}{e\rho a}}$$
$$= \sqrt{\frac{6r\varepsilon_0 m}{e\rho a}}$$

10)

If dq is one element on ring at angular position θ, then we have -

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$$F_{net} = \int_{-\pi/2}^{\pi/2} dq \times \frac{\lambda}{2\pi\in_0 R} \cdot \cos\theta \quad = \frac{\lambda q}{\pi^2 \in_0 R}$$

11) $\varepsilon = \frac{2k\lambda}{R} \sin\left(\frac{\theta}{2}\right)(\hat{i})$

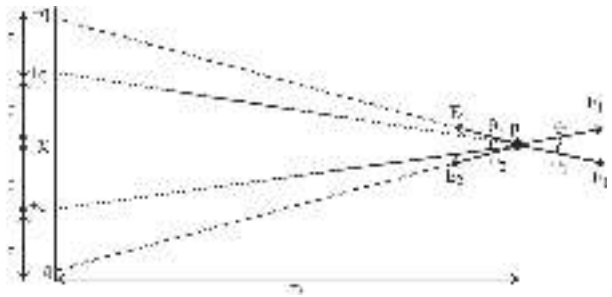
$\lambda = \left(\frac{-Q}{R\theta}\right) = \left(\frac{-Q}{R.\frac{2\pi}{3}}\right)$

$\lambda = \frac{-3Q}{2\pi R}$

$\varepsilon = \frac{2k}{R} \cdot \frac{-3Q}{2\pi R} . \sin(60°)(-\hat{i})$

$\varepsilon = \frac{3\sqrt{3}Q}{8\pi^2 \in_0 R^2}(+\hat{i})$

12) Each charge creates at point D a field intensity of $E_1 = \frac{KQ}{a^2}$. The total intensity will be the sum of three vectors (fig.). The sum of the horizontal components of these vectors will be zero, since they are equal in magnitude and form angles of 120° with each other. The vectors from angles of 90°–α with the vertical, where is the angle between the edge of the tetrahedron and the altitude h of triangle ABC.

13) 

Electric field at p = $2E_1\cos\theta_1 - 2E_1\cos\theta^2$

$= \frac{2Kq}{(d^2 + D^2)} \times \frac{D}{(d^2 + D^2)^{1/2}} - \frac{2Kq}{[(2d)^2 + D^2]} \times \frac{D}{[(2d)^2 + D^2]^{1/2}}$

$= 2KqD\left[(d^2 + D^2)^{-3/2} - (4d^2 + D^2)^{-3/2}\right]$

$= \frac{2KqD}{D^3}\left[\left(1 + \frac{d^2}{D^2}\right)^{-3/2} - \left(1 + \frac{4d^2}{D^2}\right)^{-3/2}\right]$

Applying binomial approximation Q d << D

$= \frac{2KqD}{D^3}\left[1 - \frac{3}{2}\frac{d^2}{D^2} - \left(1 - \frac{3 \times 4d^2}{2D^2}\right)\right]$

$= \frac{2KqD}{D^3}\left[\frac{12}{2}\frac{d^2}{D^2} - \frac{3}{2}\frac{d^2}{D^2}\right]$

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$$= \frac{9kqd^2}{D^4}$$

14)

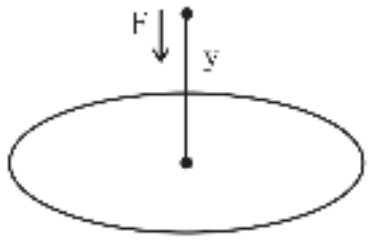
$$E = \frac{q}{4\pi\epsilon_0}\left[\frac{1}{x^2} - \frac{1}{(l-x)^2}\right]$$

at $E = 0$

$x > \frac{l}{2}$ E-towards left

$x < \frac{l}{2}$ E-towards right

15)



$$F = qE = \frac{qkQy}{\left(y^2 + R^2\right)^{\frac{3}{2}}}$$

for, $y << R$ $\quad F = \frac{qQy}{4\pi\varepsilon_0 R^3}$

Now, $\frac{qQ}{4\pi\varepsilon_0 R^3} = m\omega^2$

$$\omega = \sqrt{\frac{qQ}{4\pi\varepsilon_0 mR^3}}$$

16) $\frac{m_1}{m_2} = \frac{\sin\theta_2}{\sin\theta_1}$ $\frac{m_1}{m_2} = \frac{\sin\theta_2}{\sin\theta_1}$

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17)

According to work energy theorm

$w = \Delta T$

$W_E + W_g = \frac{1}{2}mv^2 - 0$

$W_E = qE\,l\sin\theta, \quad ...(1)$

$W_g = mg\,(l - l\cos\theta)$

$qE\,l\sin\theta + mg\,(l - l\cos\theta)$

$= \frac{1}{2}mv^2$ from eqn (1)

$mg\,l\sin\theta + mg\,l - mg\,l\cos\theta$

$= \frac{1}{2}mv^2$

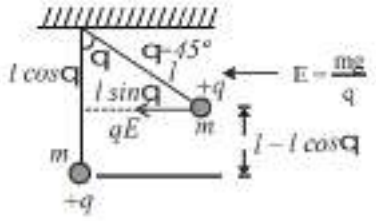
$\left\{\because E = \frac{mg}{q}\right\}$

$\frac{gl}{\sqrt{2}} + gl - \frac{gl}{\sqrt{2}} = \frac{1}{2}mv^2$

$\Rightarrow \quad v = \sqrt{2gl}$

$w = \frac{v}{l} = \frac{\sqrt{2gl}}{l} = \sqrt{\frac{2g}{l}}$

$\Rightarrow \quad \boxed{w = \sqrt{\frac{2g}{l}}}$

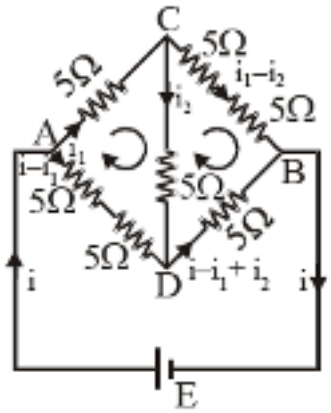


18) The given electrical circuit is an unbalanced wheat stone's bridge

i.e., $\frac{P}{Q} \neq \frac{R}{S}$

The current distribution in each branch of the bridge has been given in accordance with Kirchhoff's first law. Let us apply Kirchhoff's second law to the (a) closed mash ACDA

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$5i_1 + 5i_2 - 10(i - i_1) = 0$

or $15i_1 + 5i_2 - 10i = 0$ ....(i)

(b) closed mash CBDC

$10(i_1 - i_2) - 5(i - i_1 + i_2) - 5i_2 = 0$

or $15i_1 - 20i_2 - 5i = 0$ ....(ii)

(c) closed mash ADBEA

$10(i - i_1) + 5(i - i_1 + i_2) = E$

$-15i_1 + 5i_2 + 15i = E$ ....(iii)

Subtracting eqs. (ii) from (iii)

$25i_2 - 5i = 0 \Rightarrow i_2 = \frac{1}{5} i$ ....(iv)

Adding [4 × eq.(i) + eq.(ii)]

$75i_1 - 45i = 0 \Rightarrow i_1 = \frac{3}{5} i$

Substituting the values of i_1 and i_2 in eq. (iii), we get

$$-15 \times \frac{3}{5} i + 5 \times \frac{1}{5} i + 15i = E$$

$-9i + i + 15i = E$

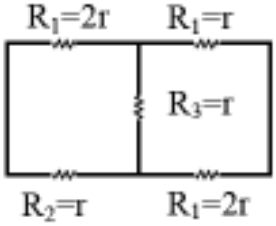
$7i = E$

If R is the equivalent resistance of the network,

then $E = Ri$ and $Ri = 7i \Rightarrow R = 7\Omega$

or

Unbalanced wheat stone Bridge Result.



$$R_{Net} = \frac{R_1R_3 + R_2R_3 + 2R_1R_2}{R_1 + R_2 + 2R_3}$$

$$= \frac{7}{5} r = 7\Omega$$

19) $$H = \frac{V^2}{R_1} t_1 = \frac{V^2}{R_2} t_2 = \frac{V^2}{R_1 + R_2} t_s = \frac{V^2}{\frac{R_1R_2}{R_1+R_2}} t_p$$

$\therefore t_s = t_1 + t_2 = 6 + 8 = 14$ min

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$$t_p = \frac{t_1 t_2}{t_1 + t_2} = \frac{6 \times 8}{6 + 8} = 3.43 \text{ min}$$

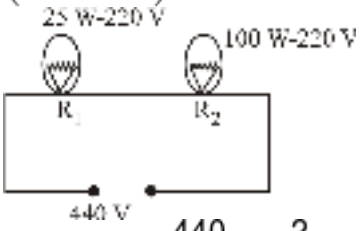
20) As

$$R_1 = \frac{220}{25} \times 220 \quad \&$$

$$R_2 = \frac{220}{100} \times 220$$

$R = R_1 + R_2 = 220 \times 220$

$$\left(\frac{1}{25} + \frac{1}{100}\right) = \frac{(220)^2}{20}\Omega$$



$$I = \frac{440}{\frac{(220)^2}{20}} = \frac{2}{11}A$$

current

Potential difference across 25W bulb

$$= IR_1 = \frac{2}{11} \times \frac{220^2}{25} = 352V$$

Potential diff. across 100W bulb

$$= IR_2 = \frac{2}{11} \times \frac{220^2}{25} = 88V$$

25W will be fused

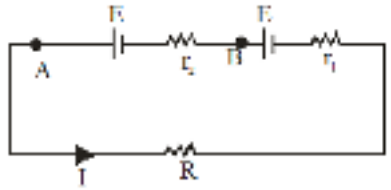
21) **Question Explanation:**

Find external resistance R such that the potential difference across the second cell (EMF E, internal resistance r_2) is zero.

Concept: Ohm's Law & Terminal Voltage

Solution:

$$I = \frac{2E}{R + r_1 + r_2} \text{ ...(i)}$$



But $V_A - V_B = E - Ir_2 = 0$

$$\Rightarrow I = \frac{E}{r_2} \text{ ... (ii)}$$

Comparing values of I from (i) and (ii)

$$\frac{E}{r_2} = \frac{2E}{R + r_1 + r_2}$$

$$\Rightarrow R = r_2 - r_1$$

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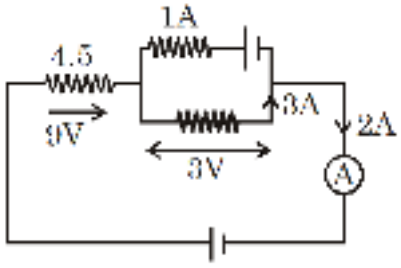
Final Answer: Option (1)

22) $E_{eq} = \frac{20 \times 10}{17} = \frac{200}{17}$

and $R_{eq} = \frac{7 \times 10}{17} = \frac{70}{17}$

23)

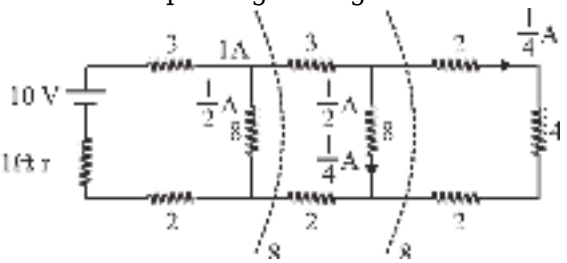
3 = 6 – i × 3
i = 1A
3V = 3 × R
R = 1Ω
E = 12V



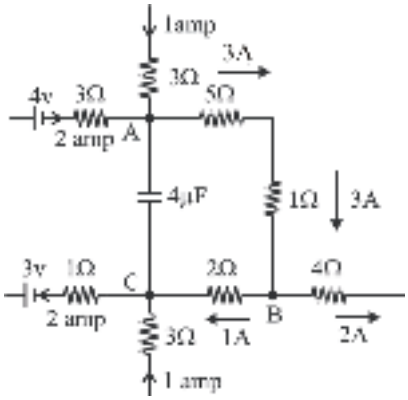
24) Equivalent resistance R_{eq} = 10 Ω so current passing through battery and 3 Ω resistance is

$i = \frac{10}{10} = A$

and current passing through 4 Ω is 0.25 A



25) Applying KCL at 'A' :-



Current through 5 Ω is = 3 A

Applying KCL at C :

Current through 2Ω is : 1A

Applying KCL at B :

Current through 4Ω is 2A

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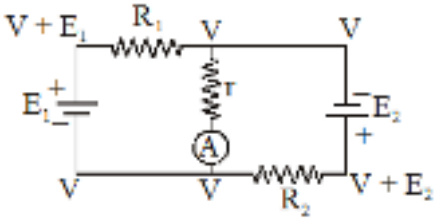
$\square U_{strored} = \frac{1}{2}C(V_A - V_C)^2$

$= \frac{1}{2} \times 4 \times 10^{-6}(15+3+2)^2$

$= \frac{1}{2} \times 4 \times 10^{-6} \times 400$

$= 800 \times 10^{-6}$

$= 800\ \mu J$



26)

No, current through r

$I = \frac{E_1}{R_1} = \frac{E_2}{R_2}$

$\frac{E_1}{E_2} = \frac{R_1}{R_2}$

27) $\frac{1}{R} = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$

$R = 3\Omega$

$i = \frac{3}{3} = 1A$

28) **Question Explanation:**

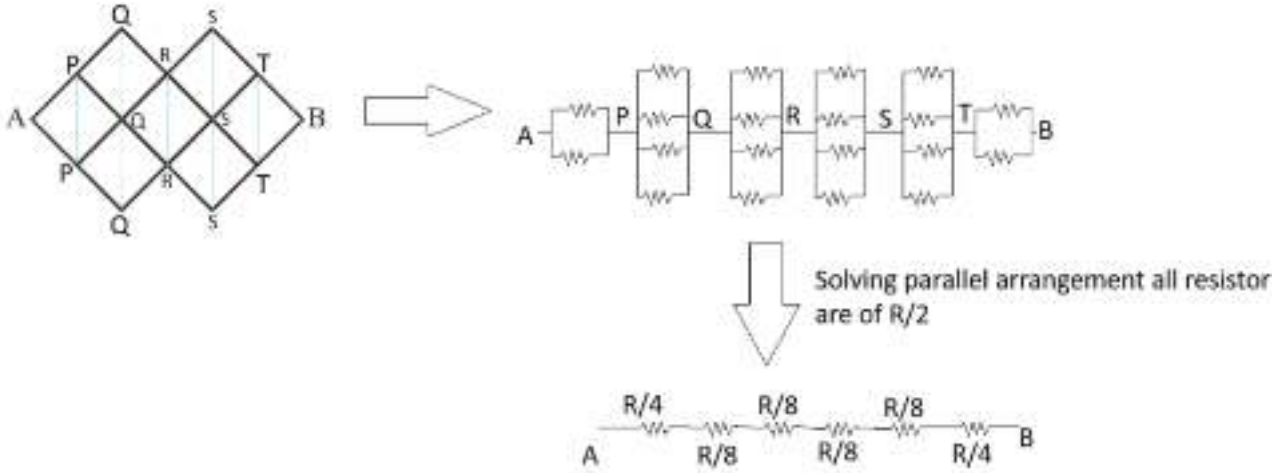
In this question , we have to determine the equivalent resistance of the circuit given

Concept:

Symmetry in circuit , Series and parallel combination of resistors

Solution:

Due to symmetry in the circuit we can determine the points at equal potential (P,Q,R,S,T) and redraw the circuit as:



$R_{Total} = \frac{R}{4} + \frac{R}{8} + \frac{R}{8} + \frac{R}{8} + \frac{R}{8} + \frac{R}{4} = R$

Final answer:

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Option (2)

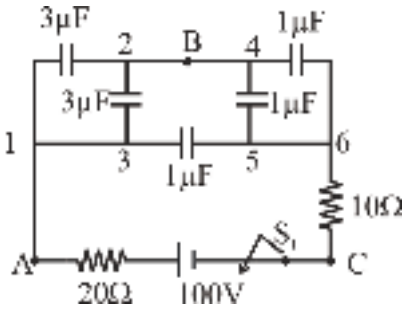
29) $\frac{1}{R} \propto P$

$\frac{1}{R_{100}} > \frac{1}{R_{60}} > \frac{1}{R_{40}}$

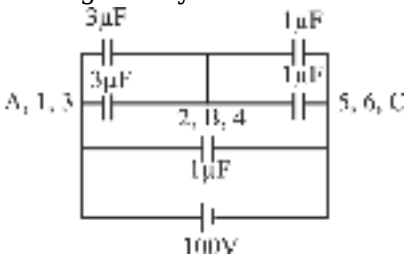
30) In steady state, the capacitor branch acts like an open circuit. So the potential difference across C is the same which is there across resistance r_2, given as

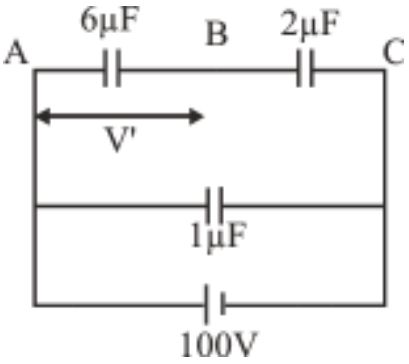
$V_C = Vr_2/(r_1 + r_2)$

31)



During steady state no current will flow hence resistors can be removed.





$V_A - V_B = \frac{100 \times 2}{8} = 25$ Volt

32) $F = Kx$

$\frac{Q^2}{2A\varepsilon_0} = K(0.2\,d)$

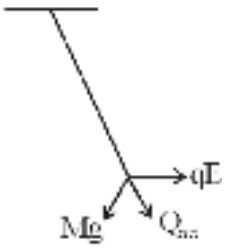
$\frac{\left(\frac{\varepsilon_0 A}{0.8d}E\right)^2}{2A\varepsilon_0} = 0.2\ Kd$

$K = 3.9\varepsilon_0 AE^2/d^3$

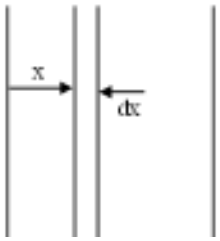
$K \simeq 4\varepsilon_0 AE^2/d^3$

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33) 

$Q_{net} \Rightarrow \sqrt{\left(\frac{qE}{m}\right)^2 + g^2}$

34) 

Taking an element of width dx at a distance x (x < d/2) from left plate

$$dc = \frac{(\varepsilon_0 + kx)A}{dx}$$

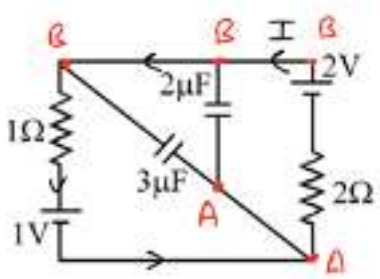
Capacitance of half of the capacitor

$$\frac{1}{C} = \int_0^{d/2} \frac{1}{dc} = \frac{1}{A}\int_0^{d/2} \frac{dx}{\varepsilon_0 + kx}$$

$$\frac{1}{C} = \frac{1}{kA} \ln\left(\frac{\varepsilon_0 + kd/2}{\varepsilon_0}\right)$$

Capacitance of second half will be same

$$C_{eq} = \frac{C}{2} = \frac{kA}{2\ln\left(\frac{2\varepsilon_0 + kd}{2\varepsilon_0}\right)}$$

35) 

In steady state

$$I = \frac{2-1}{2+1} = \frac{1}{3}$$

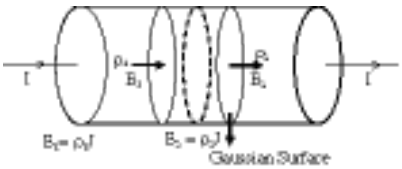
Potential difference between A and B

$$= 2 - \frac{1}{3} \times 2 = \frac{4}{3}$$

Charge on $3\mu F = \frac{4}{3} \times 3 = 4$

36) **Solution :**

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Using Gauss' Law:

$$\phi_{net} = \frac{q_{in}}{\varepsilon_0} \quad (\phi = EA)$$

$q_{in} = \varepsilon_0 (\phi_{out} - \phi_{in})$

$q_{in} = \varepsilon_0 (\rho_2 JA - \rho_1 JA)$

$(I = JA)$

$|q_{in}| = \varepsilon_0 I |\rho_1 - \rho_2|$

Final Answer : The correct option is (3)

37) **Question Explanation :** We need to find the balancing length of potentiometer in given situation.

Concept : Instrument.

Solution :

$$\frac{X}{\ell_1} = \frac{R}{100-\ell_1} \Rightarrow \frac{12}{\ell_1} = \frac{18}{100-\ell_1}$$

$\Rightarrow \ell_1 = 40$

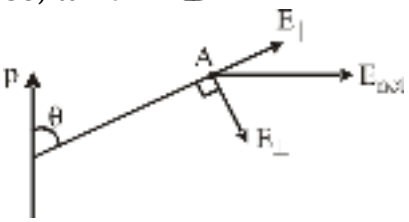
$$\frac{12}{\ell_2} = \frac{8}{100-\ell_2}$$

$\Rightarrow \ell_2 = 60$

Distance moved = $\ell_2 - \ell_1 = 60–40 = 20$ cm

Final Answer : Option (2)

38) $\tan\theta = \frac{E_{||}}{E_{\perp}}$



$\tan\theta = 2\cot\theta$

$\theta = \tan^{-1}\sqrt{2}$

39) **Question Explanation:**

Calculating electric flux through a tilted square surface.

Concept:

Electric Flux ($\phi = E \cdot A$)

Solution:

Electric flux is the dot product of the electric field vector and the area vector.

The area vector is perpendicular to the surface.

Since the electric field is only in the x-direction, only the component of the area projected onto the yz-plane (perpendicular to the x-axis) contributes to the flux.

Electric Field: $E = E_0\hat{i}$

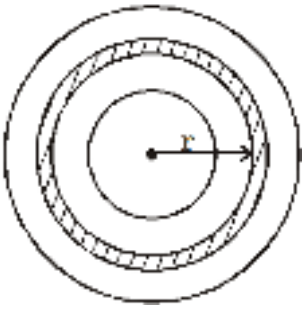
Area Projection: The square is tilted, but its projection on the yz-plane is a square of side a

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(from z = 0 to z = a and y = 0 to y = a).
Area Vector Component: The component of the area vector along the x-axis is $A_x = a \times a = a^2$.
Flux: $\phi = E_xA_x = E_0(a^2)$
$\phi = E_0a^2$
Final answer:
Option (3)

40) **Question Explanation :** The question asks to find the interaction potential energy between two parallel dipoles.
Concept : Electrostatic Potential Energy of Dipole-Dipole Interatction.
Solution :
$U = -\vec{P_2} \cdot \vec{E_{21}} = -P_2\left(\frac{1}{4\pi\varepsilon_0}\frac{P_1}{x^3}\right)\cos 180° \quad U = \frac{1}{4\pi\varepsilon_0}\left(\frac{P_1P_2}{x^3}\right)$
Final Answer : Option (2)

41) **Question Explanation:** The question asks to find b for a constant electric field.
Concept: Gauss's Law in Spherical Coordinates.
Solution:



Using guass law
$\int \vec{E}.\vec{ds} = \frac{q_{in}}{\epsilon_0}$ .....(1)
$q_{in} = q + \int_{R_1}^{r} \frac{b}{r} 4\pi r^2 dr$
$= q + 4\pi b \frac{\left(r^2 - R_1^2\right)}{2}$
from equation -1
$E\,4\pi r^2 = \frac{1}{\epsilon_0}\left(q + \frac{4\pi b\left(r^2 - R_1^2\right)}{2}\right)$
$E = \frac{1}{4\pi\epsilon_0}\left(\frac{\left(q - 2\pi bR_1^2\right)}{r^2} + 2\pi b\right)$
as $\vec{E}$ is constant, therefore
$q - 2\pi bR_1^2 = 0$
$b = \frac{q}{2\pi R_1^2} = \frac{16 \times 10^{-6}}{2\pi \times \frac{4}{\pi} \times 10^{-6}} = 2$

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Final Answer: Options (2)

42)

Question Explanation: We need to find the correct relationship between the electric flux ϕ_1 through surface S_1 and ϕ_2 through surface S_2.

Concept: Electric Flux and Solid Angle in Electrostatics

Solution:
Solid angle (Ω) subtended by both surfaces S_1 and S_2 at charge q is identical
Electric flux through any surface bounding a solid angle Ω is given by:

$$\phi = \frac{q}{\varepsilon_0}\left(\frac{\Omega}{4\pi}\right)$$

For surface $S_1 : \phi_1 = \frac{q\Omega}{4\pi\varepsilon_0}$

For surface $S_2 : \phi_2 = \frac{q\Omega}{4\pi\varepsilon_0}$

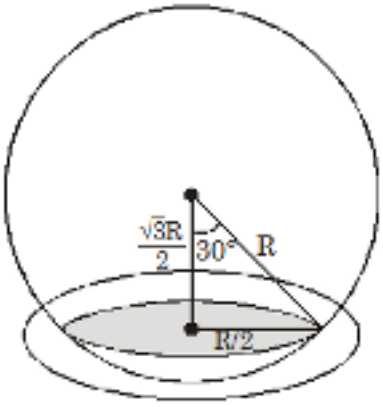
Therefore: $\phi_1 = \phi_2$

Final Answer: Option (1)

43) **Question Explanation:** Calculating electric flux through a sphere by determining the charge enclosed from a non-uniform charged disc.
Concept: Gauss's Law and Flux Calculation.
Solution: According to Gauss's Law, the net electric flux through any closed surface is $\Phi = \frac{Q_{enc}}{\varepsilon_0}$. We must find the portion of the charged disc that lies inside the sphere.
Step-by-step Calculation:

A. **Identify the Intersection:** * The disc lies in the xz-plane (y=0).
B. The sphere is centered at $(0, \frac{\sqrt{3}R}{2}, 0)$ with radius R.
C. To find the intersection, set y=0 in the sphere's equation: $x^2 + (y - \frac{\sqrt{3}R}{2})^2 + z^2 = R^2$.
D. $x^2 + z^2 + \frac{3R^2}{4} = R^2 \Rightarrow x^2 + z^2 = \frac{R^2}{4}$.
E. This means the sphere encloses a smaller circular region of the disc with radius $r_{max} = \frac{R}{2}$.
F. **Calculate Enclosed Charge (Q_{encl}):**
G. **Flux (Φ):**

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$$q_{en} = \int_0^{R/2} \sigma_0 \left(1 - \frac{r}{R}\right) 2\pi r dr$$

$$= 2\pi\sigma_0 \int^{R/2} \left(r - \frac{r^2}{R}\right) dr$$

$$= 2\pi\sigma_0 \left[\frac{R^2}{8} - \frac{R^2}{24}\right]$$

$$= \frac{2\pi\sigma_0 R^2}{6}$$

$$\phi = \frac{Q_{encl}}{\epsilon_0} = \frac{\pi\sigma_0 R^2}{6\epsilon_0}$$

Final Answer: (Option 3)

44) **Question Explanation:**

A circuit contains a capacitor where one plate is fixed and the other is attached to an elastic string (natural length ℓ, Young's modulus Y, area A). The EMF (ε) is slowly increased from zero to E_0. We need to identify the correct statement regarding the physical changes in the system.

Concept:

As voltage increases, electrostatic attraction pulls plates closer, stretching the string.

Solution:

Capacitance Change: As the EMF increases, the charges +Q and -Q on the plates increase, creating an attractive force $F_e = \frac{Q^2}{2\epsilon A}$.

This force pulls the movable plate toward the fixed plate, decreasing the separation d.

Since $C = \frac{\epsilon A}{d}$, a decrease in d leads to an increase in capacitance C. Statement 1 is incorrect.

Energy Comparison: Because the plates move closer together, the work done by the electrostatic force is stored as potential energy in the string and leads to a higher capacitance compared to a fixed-plate scenario. This results in more energy being stored in the capacitor for the same final voltage E_0. Statement 3 is correct.

Linearity of Extension: The electrostatic force is proportional to the square of the charge (and thus the square of the voltage), while the restoring force of the string is linear $(F = kx)$. Therefore, the extension does not increase linearly with the EMF. Statement 4 is incorrect.

Final answer: Option (3)

45) **Question Explanation:** Field vector of a point charge.

Concept: Coulomb's Law in Vector Form.

Solution: First, find the displacement vector r from the charge at P(3,2,5) to point S(0,–4,3) :

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$\vec{r} = \vec{r_S} - \vec{r_P} = (0-3)\hat{i} + (-4-2)\hat{j} + (3-5)\hat{k} = -3\hat{i} - 6\hat{j} - 2\hat{k}$

The magnitude is $r = \sqrt{(-3)^2 + (-6)^2 + (-2)^2} = \sqrt{9+36+4} = 7$.

Using the vector formula for Electric Field:

$$\vec{E} = \frac{kQ}{r^3}\vec{r} = \frac{9 \times 10^9 \times 10^{-6}}{7^3}(-3\hat{i} - 6\hat{j} - 2\hat{k})$$

$$\vec{E} = \frac{9 \times 10^3}{343}[-(3\hat{i} + 6\hat{j} + 2\hat{k})] = \frac{-9}{343}(3\hat{i} + 6\hat{j} + 2\hat{k}) \times 10^3 \text{ N/C}$$

Final Answer: Option (1)

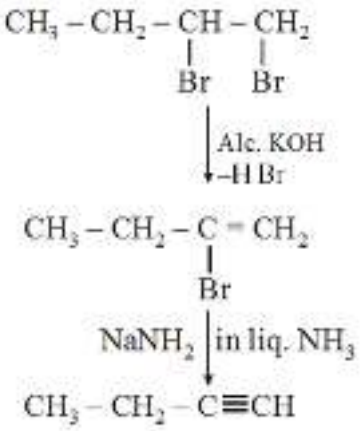
CHEMISTRY

46)

Explaining the Question :- To find out major product of the given reaction.

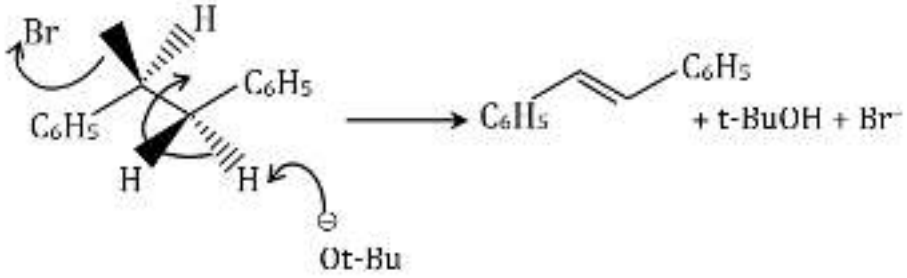
Concept:- This question is based on E-2 Bimolecular elimination reaction.

Solution :-



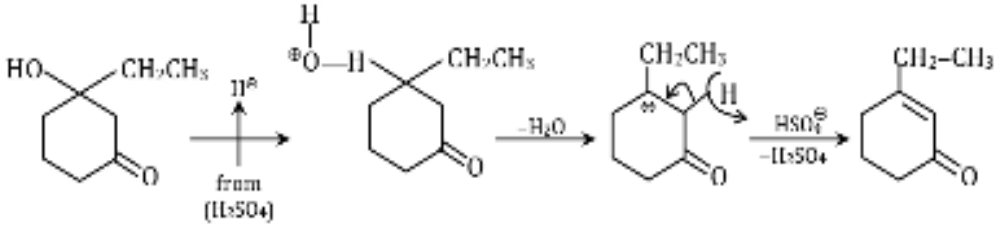
Final Answer :- 1

47) Elimination reaction is highly favoured if
(a) Bulkier base is used
(b) Higher temperature is used
Hence in given reaction biomolecular elimination reaction provides major product.

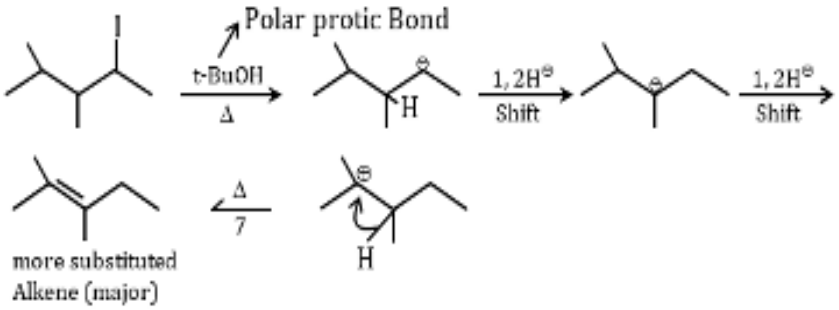


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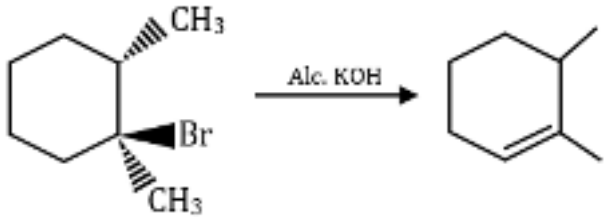
48)



49)



50)



(β-Hydrogen is present w.r.t. Br)

51) **Explanation Question:** Question is asking about the most reactive substrate towards S_N2 reaction.

Concept: This question is based on mechanism of S_N2.

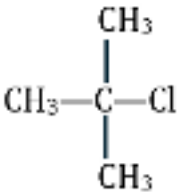
Solution: Rate of S_N2 $\propto$ Electrophilic nature of α-carbon atom of R–X

$$\propto \frac{1}{\text{Steric hinderance}}$$

$\propto$ stability of transition state

–I and –M groups increase electrophilicity of α-Carbon atom of R–X.

Over all reactivity towards S_N2 reaction is :



$O=CH-CH_2-Cl$ (–I) > CH_3-Cl (least hindered) > $CH_2=CH-CH_2-Cl$ > [most hindered]

Final Answer : Option 3

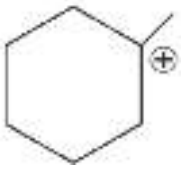
52)

Question Explanation: Comparing the rate of S_N1 reaction on hydrolysis.

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Concept: Carbocation stability.

Solution: The rate of an SN1 reaction is determined by the stability of the carbocation formed after the leaving group departs. A more stable carbocation leads to a faster reaction. The compound in option B forms a tertiary carbocation, which is the most stable among the given options due to the inductive effect of the alkyl groups.

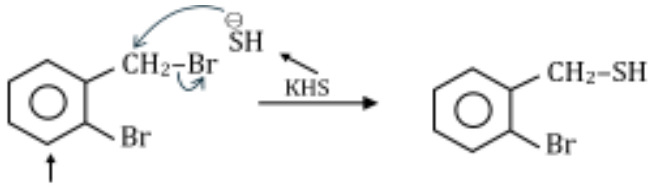


3o Carbocation is most stable.

Final Answer: Option (2)

53)

Question Explanation: Question is asking about the major product of following reaction.
Concept: This question is based on bimolecular nucleophilic substitution reaction.
Solution:

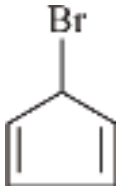


Not gives S_N2 reaction because partial double bond character between carbon-bromine bond.

v

Find answer: 4

54)

Compound 

produces antiaromatic carbocation. Hence, S_N1 is not possible.

55)

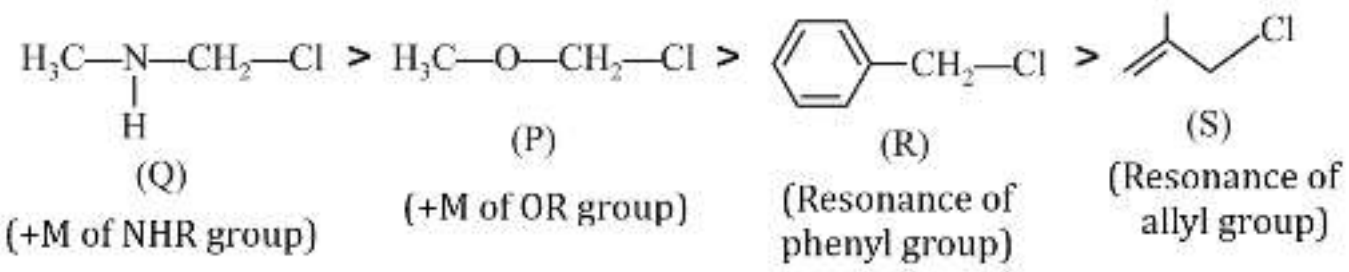
Explanation:
The question asks about the factors that affect the rate of an **S_N1 reaction**.

Concept:
S_N1 reactions proceed through a **carbocation intermediate**, so anything that stabilizes the carbocation increases the reaction rate.

Solution:

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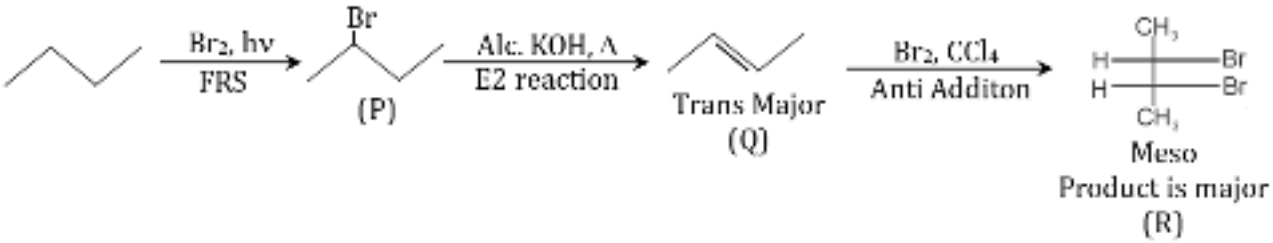
The rate of an S_N1 reaction ∝ **stability of the carbocation**.
Electron-donating groups (+I, +M) attached to the α-carbon help stabilize the carbocation and **increase the S_N1 rate**.



Final Answer: Option (3)

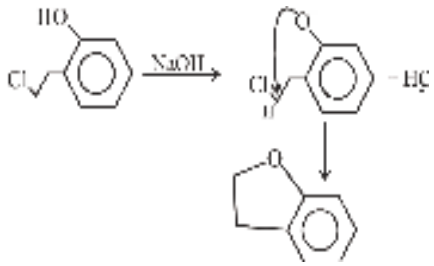
56) **Question Explain:** Question is asking about the final product **Concept:** This question is based on → Free radical substitution
→ E2 elimination
→ Electrophilic addition of Br_2

Solution:



Final Answer: Option (2)

57)



58)

Question Explanation:
The question asks why the following S_N2 reaction does not occur:

$$Br + CH_3OH \rightarrow CH_3Br + OH$$

Concept:
Leaving Group Ability, Basicity of Leaving Group, S_N2 Reaction
Solution:
For an S_N2 reaction to occur, the leaving group must be stable after departure. In the given reaction, OH^- would have to leave from methanol. Since OH^- is a strong base and a poor leaving group, the reaction does not proceed. Alcohols become good substrates for substitution only after converting the -OH group into a better leaving group (e.g., H_2O,OTs, or halides).

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Final Answer:
Option (2)

59) **Question Explanation:**

The question asks for the major product when o-bromobenzyl bromide reacts with NH_3.

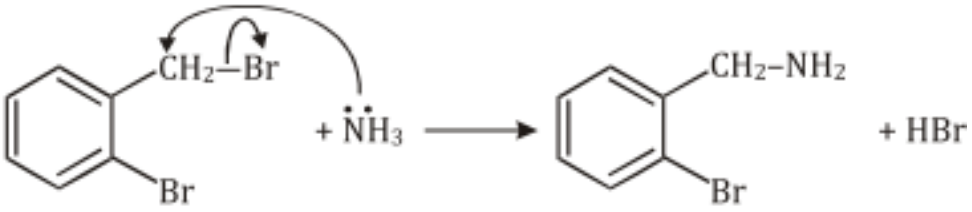
Concept:

S_N2 Reaction, Reactivity of Benzyl Halides, Aryl Halides

Solution:

Ammonia reacts with the benzylic bromide ($-CH_2Br$) via an S_N2 mechanism, replacing it with $-CH_2NH_2$. The aryl bromide (Br directly attached to the benzene ring) does not undergo nucleophilic substitution under these conditions.

Hence, only the benzylic bromine is substituted.



Final Answer:

Option (1)

60) **Question Explanation:**

The question asks for the product formed when the given cyclohexyl bromide undergoes an S_N2 reaction with OH^-.

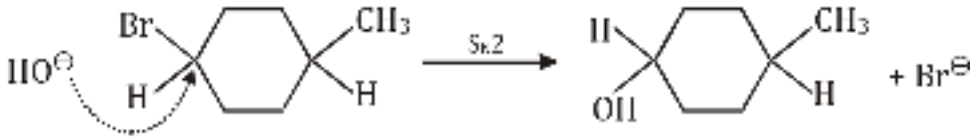
Concept:

S_N2 Reaction, Backside Attack, Walden Inversion

Solution:

In an S_N2 reaction, the nucleophile attacks from the backside of the carbon bearing the leaving group, causing inversion of configuration at that carbon.

Initially, Br is above the plane and H is below. After substitution, OH occupies the position opposite to Br, i.e., OH is below the plane and H is above.



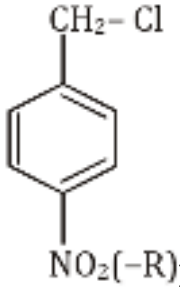
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Hence, the product corresponds to Option (2).

Final Answer:

Option (2)

61) In S_N2 Transition state, has partial carbanion character.
The stability of carbanion ∝ – R/ – H/ – I
Order of –I = $-NO_2$ > $-OCH_3$



has more stable T.S, so more reactive towards S_N2

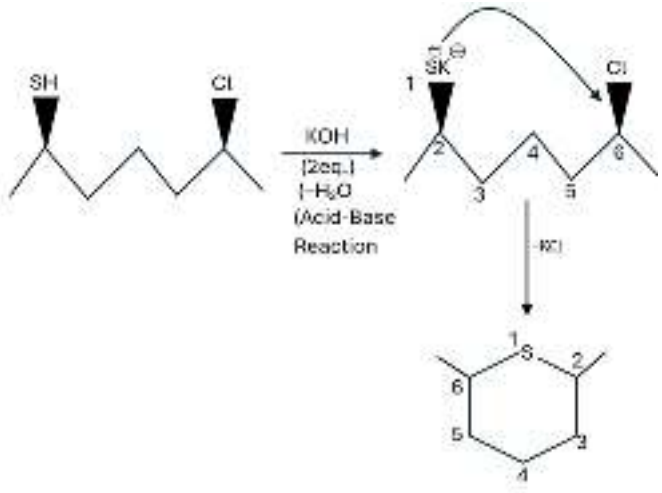
62) Via stable carbocation

63)

Explanation Question: Question is asking about the major product of above reaction.
Concept: This question is based on

A. Acid-Base reaction
B. Intramolecular nucleophilic substitution reaction

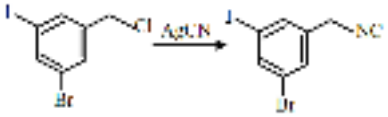
Solution :



Final Answer

The correct option is (4)

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64) 

65)

Explanation of Question: Question is asking about the compound which gives same major product by S_N^1 and S_N^2

Concept: This question is based on S_N^1 and S_N^2

Solution: In option (1), (2) and (4) carbocation show rearrangement so S_N^1 and S_N^2 products are different.
Only option (3) gives same product via S_N^1 and S_N^2 reaction because no rearrangement occurs in benzylic carbocation.

Final Answer: The correct option is (3).

66)

Question Explanation: The question asks for the correct decreasing order of the ratio between elimination (E_2) and nucleophilic substitution (S_n2) reactions for different alkyl bromides.
Concept: This question is based on comparison of S_n2 Reaction, E_2 Reaction,.

Solution : The E_2/S_n2 ratio depends mainly on the steric hindrance and structural features of the alkyl halide.

A. S_n2 reactions require backside attack by the nucleophile, so their rate decreases as steric hindrance increases.
   Order: Methyl > 1° > 2° > 3°

B. E_2 reactions are favored with more substituted alkyl halides because they form more stable, substituted alkenes.
   Order: 3° > 2° > 1°; methyl does not undergo E_2 (no β-hydrogen).

P (Tertiary, tert-butyl bromide):
Extremely hindered → S_n2 almost impossible → E_2 strongly dominates.
Highest E_2/S_n2 ratio.

A. Q (Secondary, 3-bromopentane):
   Moderately hindered → Both S_n2 and E_2 occur, but E_2 is significant.
   High but lower ratio than P.

B. S (Primary, ethyl bromide):
   Low hindrance → S_n2 is major → E_2 minor.

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Low E_2/S_n2 ratio.

C. R (Methyl bromide):
No β-hydrogen → E_2 impossible → only S_n2 occurs.
E_2/S_n2 ratio = zero → lowest.

Thus the decreasing order: P > Q > S > R
Final Answer: Option 1 is correct.

67)

Explanation:

The question asks for the major product of the reaction between phthalimide and 1-bromomethyl-4-chloromethylbenzene in the presence of KOH.

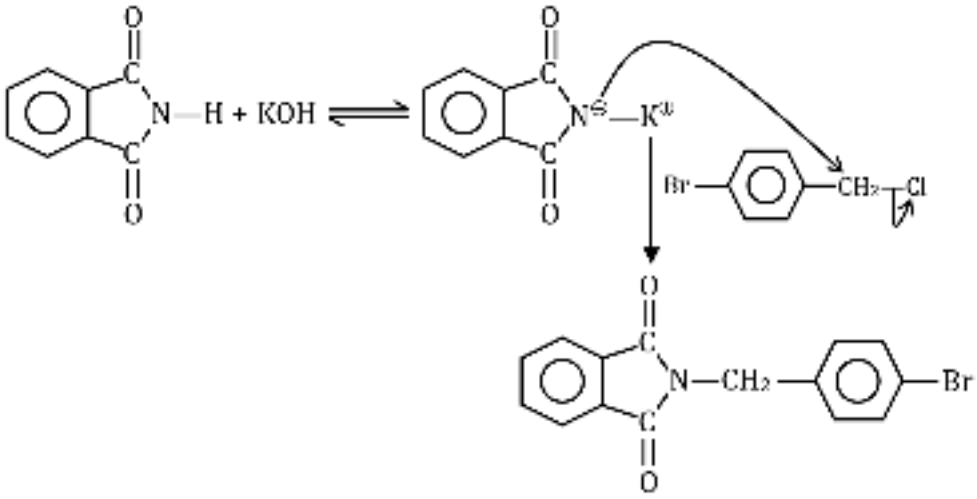
Concept:

This question is based on Gabriel phthalimide synthesis:

Step 1: Acid-Base reaction

Step 2: S_N2 Reaction (Nucleophilic Substitution)

Solution:



The nucleophile attacks the CH_2 carbon connected to the Cl atom. This is because the C−Cl bond is a primary alkyl halide and is highly reactive towards S_N2 substitution, whereas the C−Br bond is attached directly to the aromatic ring and is inert to S_N2 reaction due to partial double bond character.

The final product is N-(4-Bromobenzyl)phthalimide.

Final Answer: Option (1)

68) **Question Explanation:**

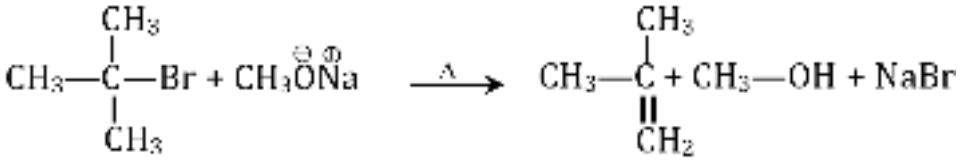
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The question asks which reaction gives elimination as the major product.

Concept:

S_N2 vs E2 Reactions, Steric Hindrance, Strong Base

Solution:



A. (1) tert-Butoxide with CH_3Br gives S_N2 (Williamson ether synthesis) because methyl halides cannot undergo elimination.

B. (2) Ethyl bromide + NaCN gives S_N2 substitution.

C. (3) Ethyl bromide + NaI gives Finkelstein reaction (S_N2).

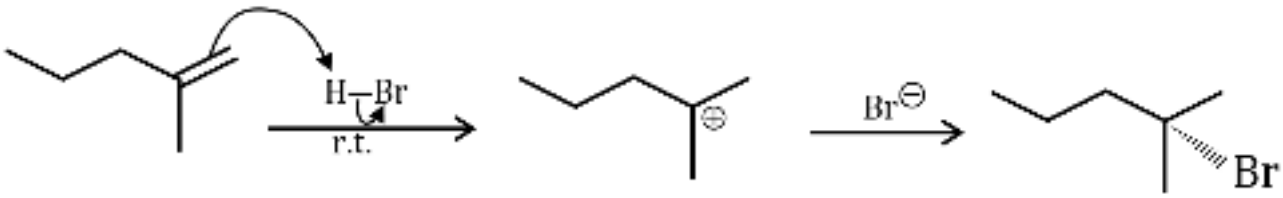
D. (4) tert-Butyl bromide with CH_3ONa and heat undergoes E2 elimination because the substrate is tertiary, making substitution difficult.

Therefore, elimination is the major reaction only in (4).

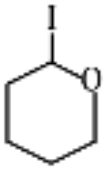
Final Answer:

Option (4)

69)



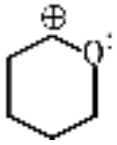
⇒ (C) HBr

70) (A)  is most reactive towards S_N1

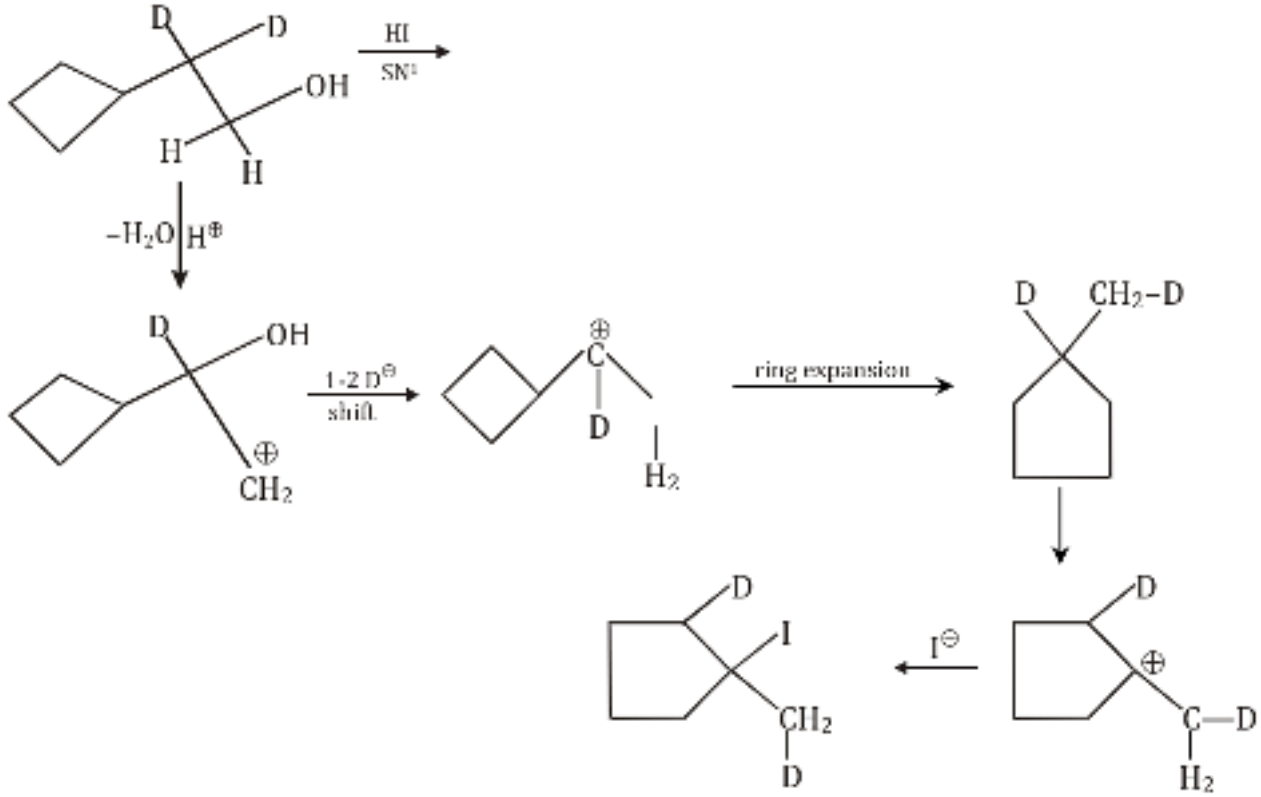
Because $I^{\ominus}$ is better leaving group so easily gives carbocation and that is more stable due to resonance

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NOVA @examlix

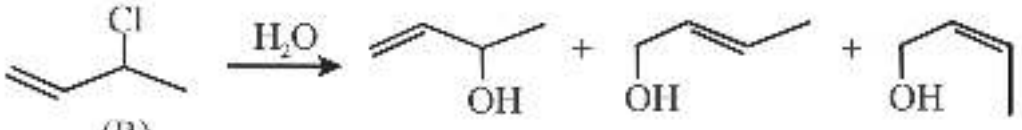
(pπ – pπ) or back-bonding. 

71)



72)

Question Explanation : What is correct for the above given reaction:
Concept: Nucleophilic Substitution reaction.
Solution: The hydrolysis of (R)-3-chloro-1-butene with water proceeds via an S_N1 and S_N1' mechanism.



(R)
optically active

Racemic mixture
(d+l)
2 products
Product mixture is optically inactive
due to external compensation

Final Answer: Option (3)

73) **Question Explanation:**
Determine the decreasing order of enthalpies (ΔH) required to form carbocations in reactions (I)–(IV).

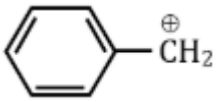
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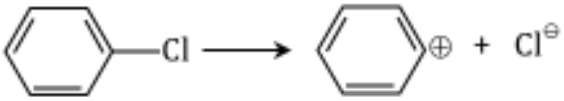
Concept:

This question is based on bond energy and carbocation stability.

A. Higher ΔH → Strong C–Cl bond or less stable carbocation formed.
B. Lower ΔH → More stable carbocation (e.g., resonance-stabilized benzylic carbocation).

Solution:

A.  Benzylic and resonance-stabilized carbocations require the least energy for formation.

B. 

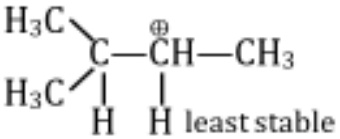
Aryl C–Cl bonds (double bond character) are exceptionally strong → require the highest energy to break.

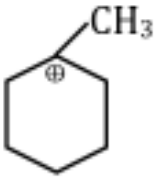
Stability order of formed carbocations: 3 > 2 > 1 > 4.

Hence, decreasing order of enthalpies: $\Delta H_4^\circ > \Delta H_1^\circ > \Delta H_2^\circ > \Delta H_3^\circ$

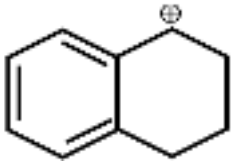
Final Answer: Option (2)

74) Reactivity towards Solvolysis in aqueous medium ∝ stability of carbocation

(i)  least stable

(ii) 

(iii) $H_3C_6—\overset{\oplus}{C}(H)—C_6H_5$ most stable

(iv) 

decreasing order of stability III > IV > II > I

75) **Question Explanation:**
Which organic bromide forms an AgBr precipitate upon ionization in aqueous $AgNO_3$?

Concept:

This question is based on carbocation stability, aromaticity in S_N1 reactions.

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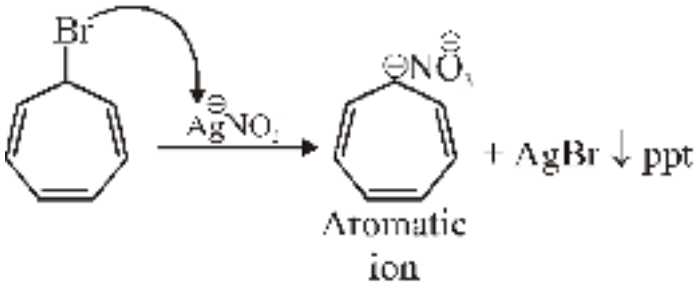
Solution:

A. Precipitation: AgBr precipitate forms if the C-Br bond ionizes easily to release Br^-.

B. Unreactive Options (1, 2, 3): Br is on unreactive sp^2 (vinyl/aryl/pyridyl) due to partial double bond character between C–Br bond.

C. Reactive Option (4): Tropyl bromide ionizes to the highly stable, aromatic Tropylium ion.

D. Conclusion: The formation of the stable aromatic carbocation drives the release of Br^-, leading to AgBr precipitation.



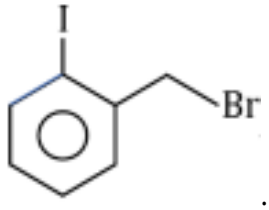
Final Answer: Option (4)

76)

Explanation of Question:
(A) Question is asking about the product (B) in the given reaction

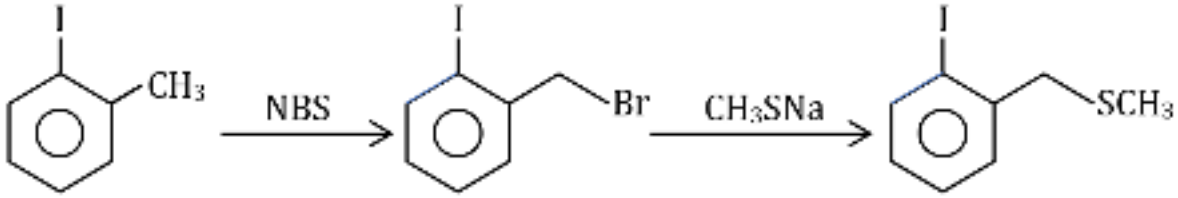
Concept: This question is based on reaction of NBS and S_N^2 reaction mechanism.

Solution: NBS gives free radical substitution reaction at allylic and benzylic position will give



(A) .

CH_3SNa will substitute Br in (A) by S_N2 mechanism

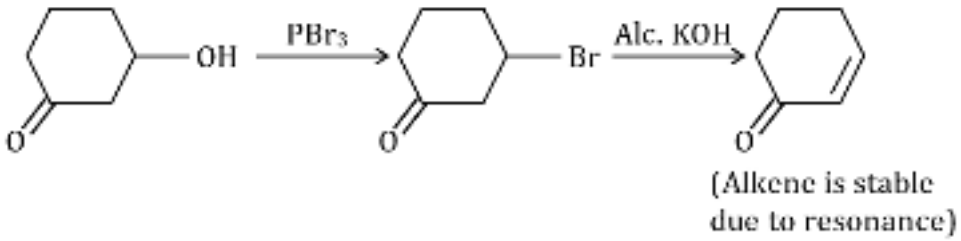


Final Answer:
The correct option is (1)

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77) **Question Explanation:** Question is asking to determine the major product of the reaction sequence of 3-hydroxycyclohexanone with PBr_3 followed by alcoholic KOH.
Concept: This question is based on multiple concept
(1) Nucleophilic substitution of alcohol
(2) E2 elimination reaction

Solution:



Final Answer : Option (4)

78) **Question Explanation:**

The question asks for the major product formed when the given primary alkyl bromide reacts with CH_3O^-/CH_3OH.

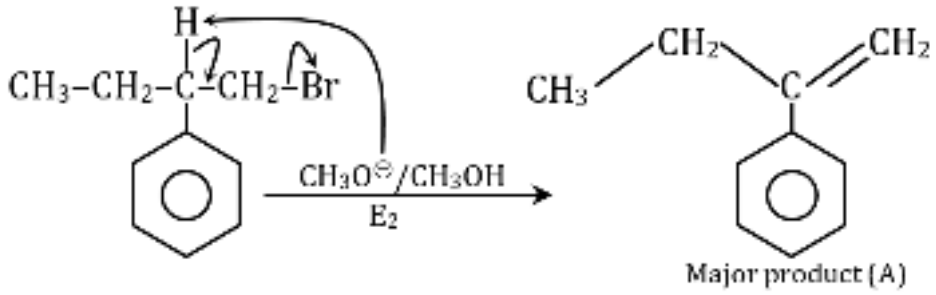
Concept:

E2 Elimination, Strong Base, Saytzeff Rule

Solution:

Methoxide ion (CH_3O^-) is a strong base. It abstracts a β-hydrogen, while Br^- leaves simultaneously via an E2 mechanism. Elimination forms the more substituted alkene, which is conjugated with the phenyl ring and hence more stable.

The major product is:



Final Answer:

Option (2)

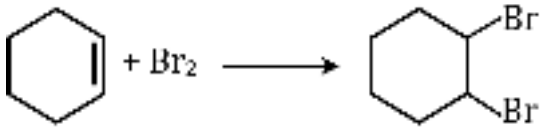
79) (a) $CH_3CH_2CH_2Br + KOH \longrightarrow CH_3CH{=}CH_2 + KBr + H_2O$

(b)



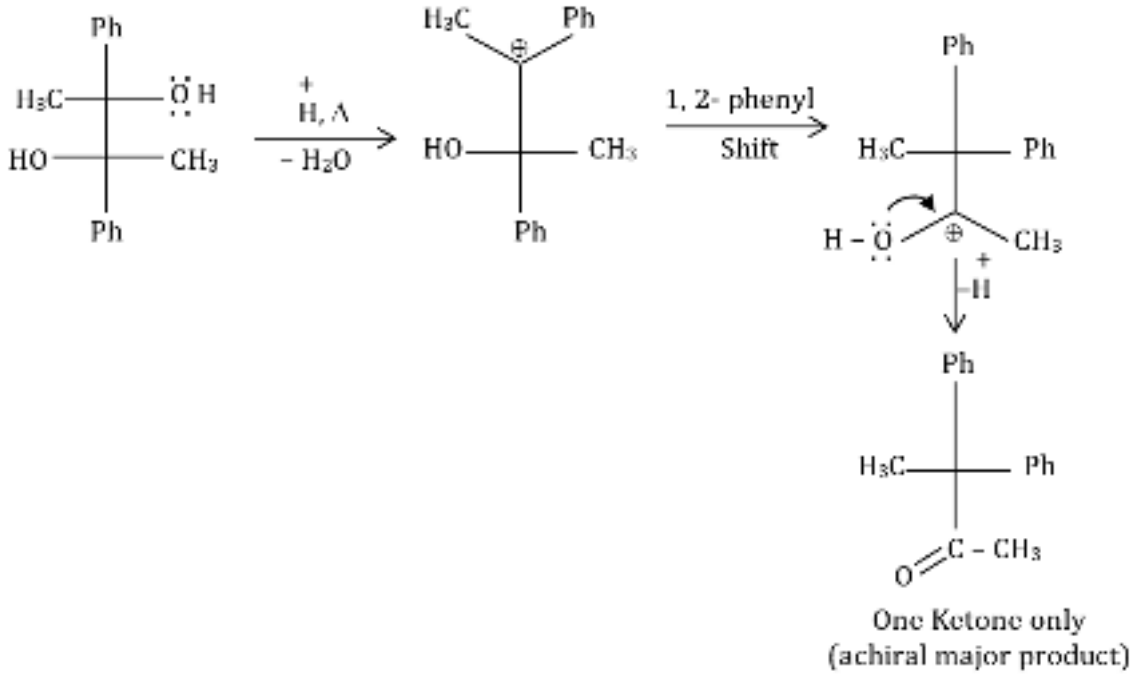
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(c)



(a) → Elimination reaction
(b) → Substitution reaction
(c) → Addition reaction

80) **Question Explanation:** Asking about the number of Pinacolone products.
Concept : This question is based on Pinacol - Pinacolone rearrangement.
Solution: Pinacol - Pinacolone rearrangement of vicinal diols involve intermediate carbocation and rearrangement of carbocation.
Mechanism is similar to E⊥ elimination.



Final Answer

The correct option is (4)

81)

Question Explanation:
The question involves formation of a ketone through a carbocation intermediate where a phenyl group migrates preferentially.

Concept :
This question is based on Semi-Pinacol Rearrangement, Carbocation Rearrangement, Migratory Aptitude.

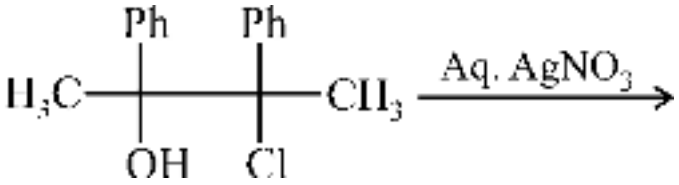
Solution:

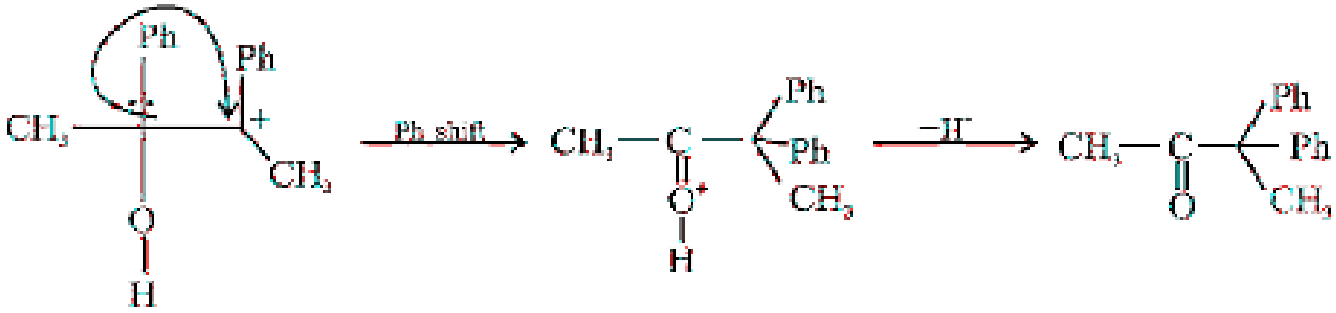
A. Ag^+ ions from $AgNO_3$ remove Cl^- as AgCl(s), generating a tertiary carbocation at the carbon bearing chlorine.

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B. A 1,2-shift then occurs from the adjacent carbon. Since phenyl (Ph) has higher migratory aptitude than methyl (CH_3), the phenyl group migrates.

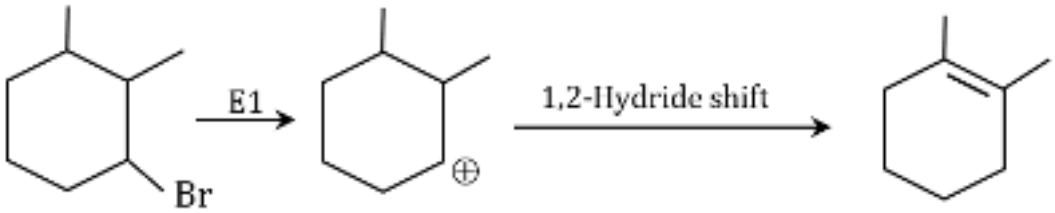
The adjacent –OH group loses H^+, forming a stable C=O (ketone).





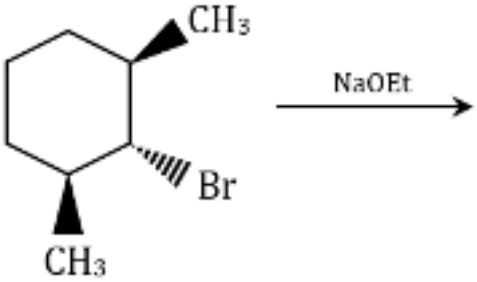
Final answer : Option (2)

82) **Explain Question:** To find product formed in E1 reaction of 1 – Bromo -2, 3 – dimethylcyclohexane
Concept: E_1 reaction.
Solution:



Final Answer: (D)

83) **Explain Question:** To find product of given reaction.
Concept: E2 Reaction
Solution:



No product [No b-hydrogen on wedge w.r.t Br]

Final Answer: (D)

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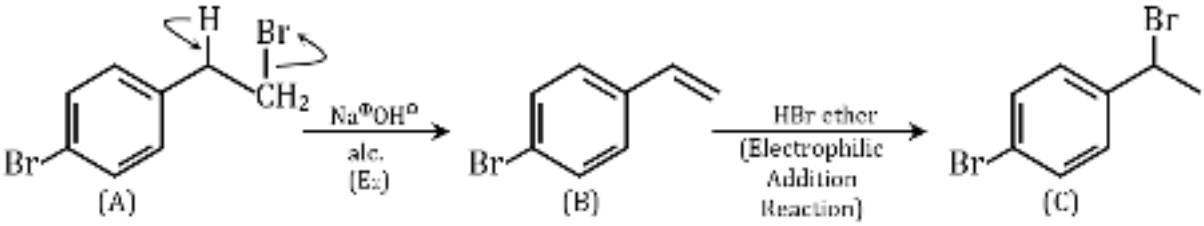
84) Decreasing rate of E_2 reaction :

R > P > Q

R: ↓ β-hydrogen on dash w.r.t Br

Q: ↓ No β-hydrogen on wedge w.r.t Br

85)



A and C are position isomer.

86) **Explain the question:** Identify major/minor alkenes and elimination type.
Concept: Hofmann elimination via E2 mechanism with E1cb character

Solution:

$$CH_3CH_2CH_2\underset{\oplus NMe_3}{CH}-CH_3 \xrightarrow[EtOH]{EtO^-} \underset{(A)\ (major)}{CH_3CH_2CH_2CH=CH_2} + \underset{(B)\ (minor)}{CH_3CH_2CH=CH-CH_3}$$

Elimination of quaternary ammonium salts $(-N(CH_3)_3)$ using a strong base like C_2H_5ONa in C_2H_5OH follows the Hofmann rule due to the bulky leaving group and carbanion-like transition state. Deprotonation preferentially occurs at the less substituted β-carbon (terminal methyl), making the less substituted alkene, pent-1-ene ($CH_3CH_2CH_2CH=CH_2$), the major product A, while pent-2-ene ($CH_3CH_2CH=CHCH_3$) is the minor product B. Since the less substituted alkene is the major product, these are classified as Hofmann products.

Final Answer: Option (3)

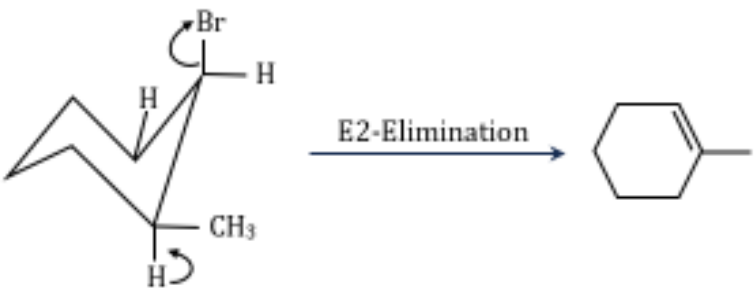
87) **Explain Question:** To find alkyl halide which is least reactive in E1 reaction
Concept: E1 reaction
Solution:

$CH_2=\overset{\oplus}{C}-CH_2-CH_3 \rightarrow$ Least stable intermediate therefore, least reactive alkyl halide among given option for E1 reaction is

$$CH_2=\underset{\displaystyle Br}{\underset{|}{C}}-CH_2-CH_3$$

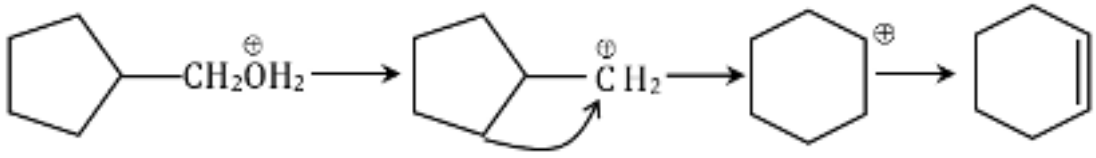
Final Answer: (C)

88)

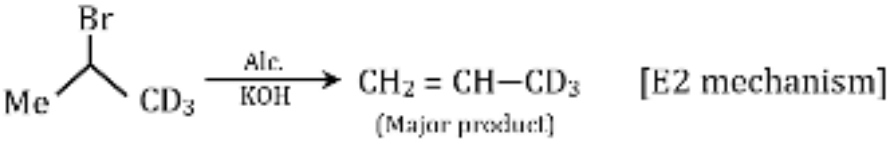


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89)



90)



BIOLOGY

91)

Question Explanation: The question is based on the principle of agarose gel electrophoresis.

Concept: Gel electrophoresis.

Solution: DNA fragments move toward the anode (positive pole) because DNA is negatively charged.Smaller fragments move faster and farther, while larger fragments move slower and stay closer to the wells/cathode.

So ,Band 1 is closest to the well/cathode → it must be the largest DNA fragment.Bands 2 and 3 (farther away) are smaller.

Final answer: Option 3

92) **Explanation :**
How many of the given statements regarding biotechnology techniques are correct.

Concept :
The concept based on Tools of recombinant DNA technology.

Solution :
Statement (A) is incorrect because source of restriction enzyme hind III is Haemophilus influenza not E.coli.
Statement (B), (C) and (D) are correct.

Final Answer : (2)

93)

Question Explanation: The question asks how many statements about genetic engineering, biotechnology, and related technologies are correct.

Concept: The question is based on principles and applications of biotechnology

Solution: The correct answer is option 2.

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A. Statement (a) is correct: Genetic engineering involves techniques to directly alter DNA/RNA.

B. Statement (b) is correct: Bioprocess engineering manages conditions for growth of organisms and product formation.

C. Statement (c) is incorrect: Biotechnology is applied in medicine, industry, and environmental science, not restricted to agriculture.

D. Statement (d) is correct: Recombinant DNA technology is a major component of genetic engineering.

Final answer: Option 2

94)

Question Explanation: The question is asking which one of the given pairs correctly matches the term with its description.

Concept: Insertional inactivation and identification of recombinants

Solution: In insertional inactivation, when a gene is inserted into a selectable marker site, that marker loses its function. This helps differentiate recombinants from non-recombinants, which matches the given description correctly.

Final Answer: 1

95)

Question explanation : Identify which statements about biotechnology methods are correct.
Concept : Biotechnology techniques
Solution : The correct answer is option 2.

A. (a) Incorrect → In **microinjection**, DNA is directly injected into the cell using a fine needle. Bombardment with gold particles is the **biolistic (gene gun) method**, not microinjection.

B. (b) Incorrect → In PCR, **Taq DNA polymerase is thermostable** and does not denature at high temperature.

C. (c) Correct → A stirred-tank reactor is cylindrical or with a curved base to facilitate mixing.

D. (d) Correct → Transformation is the process of introducing foreign DNA into a host bacterium.

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Thus, only (c) and (d) are correct.

Final answer : Option 2

96)

Question Explanation: Check whether restriction enzymes link foreign DNA to plasmid and if they are called molecular glue.

Concept: Enzymes

Solution: The correct answer is option 4.

A. Assertion (A): Incorrect – Restriction enzymes do not link DNA; they cut DNA at specific recognition sites. The enzyme that joins foreign DNA with plasmid is DNA ligase.

B. Reason (R): Incorrect – Restriction enzymes are not called molecular glue. DNA ligase is called molecular glue because it joins DNA fragments.

Final Answer: Option 4

97)

Question Explanation – The question evaluates if E.coli with pBR322 insert at Pvu I site becomes ampicillin sensitive (A) and if Pvu II site is in tetR gene (R).

Concept – cloning vector

Solution –
Assertion (A) True: Pvu I site is in ampR gene. Insert at Pvu I disrupts ampicillin resistance → becomes ampicillin sensitive.
Reason (R) False: Pvu II site is in ROP gene, not tetR gene.

Final Answer – Option 3

98) **Solution:** The colonies of recombinant bacteria appear white in contrast to blue colonies of non-recombinant bacteria because of : Insertional inactivation of beta-galactosidase in recombinant bacteria
Final Answer: (4)

99) **Concept:**
Purpose of Heat Shock

Solution:
The combination of divalent cation treatment and heat shock works together to force the rDNA into the bacterium by altering the efficiency with which DNA enters through the cell

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wall/membrane pores.

A. **Why others are incorrect:** (1) DNA must remain double-stranded; (2) Selection happens on antibiotic plates, not by heat; (4) Ori activation is not temperature-dependent in this context.

Final answer: Option (3)

100) **Concept:**
Steps in DNA isolation

Solution:
First, the cell wall must be broken (B). Then, macromolecules like RNA (D) and Proteins (A) are removed by specific enzymes. Finally, the purified DNA is precipitated (C). While D and A can sometimes occur simultaneously, the logical flow starts with cell lysis and ends with precipitation.

A. **Why others are incorrect:** Any sequence starting with A or D is incorrect because the cell must be lysed (B) first. **Correct Answer:** 2 (or 1, as NCERT lists enzymatic treatments together after lysis) — *Refining for specificity:* NCERT says: "The RNA can be removed by... Ribonuclease whereas proteins can be removed by... Protease." Usually, lysis is first, precipitation is last.

Final answer: Option (2)

101) **Concept:**
Host-Vector techniques
Detailed Solution:
The incorrect statement is **Option 4**

A. **Micro-injection** – Correct. DNA is directly injected into the nucleus of animal cells using a fine needle.
B. **Biolistics** – Correct. Also called the gene gun method, it uses high-velocity micro-particles (like gold or tungsten) to deliver DNA into plant cells.
C. **Disarmed pathogens** – Correct. For example, Agrobacterium tumefaciens is modified (disarmed) and used to transfer recombinant DNA into plant cells.
D. **Competent host cells with monovalent cations** – **Incorrect.** Competent cells are typically prepared using **divalent cations** like calcium ions (Ca^{2+}), not monovalent cations like sodium (Na^{+}).

Final Answer: Option 4

102) **Concept:**
Requirement for Replication in Host

Solution:
For any alien DNA to replicate in a host, it must be integrated into the host's genome or be part of a vector that has an 'ori'. Without an 'ori', the DNA cannot multiply.

A. **Why others are incorrect:** (1) Gold is an alternative to Tungsten; (3) Ethanol is for isolation; (4) The question states biolistics was used.

Final answer: Option (2)

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103)

Explain Question : Determine why both passenger DNA and cloning vector are cut with the same restriction enzyme in recombinant DNA technology.

Concept : Compatible sticky ends

Solution : Restriction endonucleases recognize specific palindromic DNA sequences and cut DNA to produce sticky ends (single-stranded overhangs). When the same restriction enzyme cuts both the cloning vector and the passenger DNA, they generate complementary sticky ends. These ends can base-pair with each other, allowing DNA ligase to join them efficiently and form recombinant DNA.

Final Answer : Option 2. Compatible sticky ends are produced for efficient ligation

104) **Concept :** Visualization of DNA.

Solution: Ethidium bromide is the staining agent. Without it, DNA (which is transparent) cannot be seen even under UV light.

Correct Answer: The DNA bands will not be visible under UV light.

105)

Explain Question : Identify the feature of a cloning vector whose absence would prevent a recombinant plasmid containing passenger DNA from replicating inside the host cell.

Concept : Origin of replication

Solution : For a plasmid (cloning vector) to replicate inside a host cell, it must contain an origin of replication (ori). The ori is a specific DNA sequence where the replication machinery of the host initiates DNA replication. If the ori is missing or damaged, the plasmid cannot replicate, even if the passenger DNA has been successfully inserted.

Final Answer : Option 2. Ori (origin of replication)

106)

Concept :
Principle of disarmed vectors.

Solution:

A. **Correct (3):** NCERT mentions that pathogens like *Agrobacterium tumefaciens* or Retroviruses are "disarmed" so they are no longer pathogenic (don't cause disease/tumors) but are still used to deliver desirable genes into the host.
B. **Incorrect (2):** They must still be able to replicate/integrate to be effective.

Final Answer: Option (3)

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107)

Concept :
Comparison of direct gene transfer methods.

Solution:

A. **Correct (2):** Both micro-injection and biolistics (gene gun) are "direct" methods, meaning they do not rely on a biological vector like a plasmid or virus to carry the DNA into the cell.
B. **Incorrect (1):** They are typically used for eukaryotes (animals and plants respectively).
C. **Incorrect (3):** Only biolistics uses gold/tungsten.
D. **Incorrect (4):** $CaCl_2$ is for chemical competence in bacteria.

Final Answer: Option (2)

108)

Concept :
Chronology of transformation protocol.

Solution:
According to NCERT, the host is first treated with a divalent cation (Ca^{2+}).
Then the rDNA is incubated with the cells on **ice**, followed by a brief **heat shock (42˚C)**, and finally putting them back on **ice**.

Final Answer: Option (2)

109)

Concept :
Steps in DNA isolation and purification.

Solution:

A. (A) is correct; RNA is a contaminant removed by Ribonuclease.
B. (B) is correct; Ethanol must be chilled for precipitation.
C. (C) is correct; Spooling is the term used for the collection of DNA threads.
D. (D) is correct; In eukaryotes, DNA is intertwined with proteins like histones; proteases remove them to ensure the DNA is pure.

Final Answer: Option (4)

110) **Concept:**
Mechanism of Endonuclease action.

Solution:
A restriction enzyme cuts each of the two strands of the double helix.
Therefore, to cut through a linear DNA molecule at one site, it must break 2 phosphodiester bonds (one on each strand).

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Final answer: Option(2)

111) **Concept:**
Recombinant DNA technology.

Solution: While sticky ends form H-bonds spontaneously, DNA ligase is the enzyme that joins the DNA fragments by creating covalent bonds in the backbone.

Final answer: Option(3)

112) **Concept:**
Isolation of Genetic Material (Macromolecules).

Solution: DNA is not free in the cell; it is intermixed with proteins (histones), RNA, and other molecules.
These must be removed by specific treatments (proteases, etc.) to obtain pure DNA.

Final answer: Option(2)

113) **Concept:**
Sticky end formation.

Solution: The cut is between G and A: 5' - G | AATTC - 3'.
The overhanging (unpaired) part on this strand is AATTC.

Final answer: Option(2)

114) **Concept:**
Functioning of Restriction Endonucleases.

Solution: Both statements are correct lines.
However, (R) explains how the enzyme finds the site, while (A) describes the specific action once found.
In the context of "why" it cuts specifically there, the inspection is the prerequisite step.

Final answer: Option(1)

115)

Concept:
Action of Endonucleases.

Solution:
A restriction endonuclease cuts each of the two strands of the double helix at specific points in their sugar-phosphate backbones. Thus, 2 phosphodiester bonds are broken per recognition site.

Final Answer: Option (2)

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116) **Concept:**
Plant Transformation Tools.

Solution:
For plants, the Ti plasmid (from *Agrobacterium*) and Biolistics (Gene gun) are the standard NCERT-prescribed tools. Retroviruses and Micro-injection are for animals.

Final Answer: Option (2)

117) **Concept:**
Visualization of DNA.

Solution:
Ethidium bromide is the staining agent. Without it, DNA (which is transparent) cannot be seen even under UV light.

Final Answer: Option (3)

118) **Concept:**
Chromogenic Substrate Selection.

Solution:
Non-recombinants have an intact *lacZ* gene, which produces β-galactosidase. This enzyme breaks down the chromogenic substrate (X-gal) to produce a blue color. Recombinants, due to insertional inactivation, appear white/colorless.

Final Answer: Option (2)

119)

Question Explanation: The question asks which statement about pBR322 vector is incorrect.

Concept : About Question Cloning Vectors

Solution:

A. (1) Correct – pBR322 is derived from naturally occurring plasmids of E. coli.

B. (2) Correct – It has two antibiotic resistance genes: ampR and tetR.

C. (3) Correct – Developed by Bolivar and Rodriguez.

D. (4) Incorrect – pBR322 does not use colour differentiation (like blue-white screening). It uses antibiotic resistance for selection.

Final Answer: Selectable markers present in it can differentiate recombinants from non-

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recombinants on the basis of their inability to produce colour in the presence antibiotic.

120)

Question Explanation: Identify the correct statements related to recombinant DNA technology.

Concept: Biotechnology

Solution: The correct answer is option 3.

A. (a) Correct – When different DNA molecules are cut with the same restriction enzyme, they produce the same type of sticky ends, which can pair with each other.

B. (b) Correct – After agarose gel electrophoresis, the desired DNA band is cut out and the DNA is extracted from the gel. This process is called elution.

C. (c) Incorrect – RNA is removed by RNase, not by protease. Protease digests proteins.

D. (d) Correct – A protein produced by expressing a gene in a heterologous host is called a recombinant protein.

Final Answer: Option 3

121)

The correct answer is 2. a and c only.

A. (a) True: Agarose is commonly used in gel electrophoresis.

B. (b) False: Plasmids and bacteriophages can replicate in bacterial cells.

C. (c) True: Transformation introduces DNA into bacteria.

122) The correct answer is **option 4. (a), (b) and (d)**

Solution/Explanation:

Incorrect statements:

A. (a) DNA is negative, moves to the anode.
B. (b) DNA moves through the gel, and gel concentration does affect movement.
C. (d) Pure DNA needs staining to be seen under UV light.

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Correct statement:

A. (c) Smaller fragments travel father.

Hence, option (4) is correct.

123) 5' _____ GAATTC _____ 3'
3' _____ CTTAAG _____ 5'
NCERT Pg No 166, 167

124)

NCERT XII Pg # 174, fig 9.7 (a)

126)

NCERT Pg. No. 178

127)

NCERT Pg. No. 194, 200, 201

128)

XII OLD NCERT Pg # 199

129)

A - ECO R1 B - BAM H1 C - ampR D - ORI
Hence option 4 is correct

130)

Solution : Downstream processing includes:

A. (a) Quality control

B. (b) Clinical trial

C. (c) Purification

D. (e) Separation

Production of recombinant protein and maintenance of optimal growth condition part of upstream processing.

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Final Answer ; (3)

131)

Conceptual

132) Excessive agitation in sparged bioreactors can cause shear stress, damaging cells or denaturing sensitive proteins, reducing yield despite increased oxygen transfer and biomass. The document highlights stirred tanks for mixing and aeration but implies optimization to avoid such issues.

133) **Solution :** Extracting DNA bands from the gel is called elution, not spooling. Agarose is a natural polysaccharide polymer used as a gel matrix. agarose is a national polymer extracted from sea seeds.
Final Answer : (2)

134)

Question Explanation :Identify the Insertional inactivation disrupts lacZ gene so β-galactosidase is not produced.
Concept : Insertional inactivation
Solution : The correct answer is option **3**

Insertional inactivation occurs when a foreign DNA fragment into the lacZ gene coding for b–galactosidase.

A. In recombinant colonies the lacZ gene disrupted, so galactosidase is not produced.
B. When chromogenic substrate (like X-gal) is present, these recombinant colonies remain white because there is not enzyme to convert the substrate to blue.

Non-recombination colonies have an intact lacZ gene and produce blue colonies in the presence of X-gal.

Final answer : Option 3

135)

Explain Question: The question asks to analyse the given assertion and reason statements and select the correct option.

Concept: Tissue culture (Continuous Culture)

Solution:

Assertion A: For large-scale production of a recombinant protein, the cells harboring the cloned gene of interest need to be cultured in a continuous system.

Reason R: A continuous culture system allows cells to remain in their most active

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log/exponential phase by adding fresh medium and draining used medium.

Both A and R are true and R is the correct explanation of A.

So, the correct answer is: 1.

Final answer: Option 1

136) Answer : (i), (ii), (iii) and (iv)

137) **Explaining : (A)** "*Flowers are morphological and embryological marvels and the sites of sexual reproduction.*"
Correct – Flowers are the **reproductive structures** in angiosperms, and their development involves **morphological and embryological complexity**.
(B) "*Much before the actual flower is seen on a plant, the decision that the plant is going to flower has taken place.*"
Correct – **Floral induction** (the decision to flower) occurs **before** any visible signs of flowering appear.
(C) "*Flowers have always been used as symbols for conveying important human feelings.*"
Correct – Culturally and historically accurate (e.g., love, grief, celebration).
(D) "*Human beings have had an intimate relationship with flowers since time immemorial.*"
Correct – Flowers have been part of **rituals, art, religion**, etc., across civilizations.
(E) "*Several hormonal and structural changes are initiated which lead to the differentiation and further development of floral primordium.*"
Correct – **Hormones** like gibberellins and auxins trigger **floral meristem formation** and primordium development.
(F) "*Flowers are objects of aesthetic, ornamental, social, religious and cultural values.*"
Correct – True and widely accepted across all human societies.
Concept :
Final Answer : (1)

138) NCERT XII PAGE NO 17

139)

NCERT XII PAGE NO 20

140) NCERT XII P. NO 24

141) **Solution :** Exine containing sporopollenin protects pollen from adverse conditions.
Final Answer : (2)

142)

Conceptual

Watermarkly

143) NCERT (XII) Pg. # 30

146)

Conceptual

150)

Conceptual

151)

Conceptual

152)

Conceptual

153)

Conceptual

155) Conceptual

156) Conceptual

157) Conceptual

158) **Concept:** Gamete formation in dihybrid crosses
Explanation:

A. RrYy → 4 types of gametes (RY, Ry, rY, ry)
B. RRYY → RY only
C. rryy → ry only
D. RRYy → RY, Ry

Final Answer : (1)

159)

Conceptual

160)

Explanation:
Wilkins and Franklin used X-Ray Crystallography to study the structure of DNA. Franklin's

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"Photo 51" was particularly important.

Correct Answer:
Option (3)

161)

Correct Answer: 2.
Two **Short**

Explanation:
Due to semi-conservative DNA replication, the two original radioactive strands will each be retained in a separate DNA molecule after any number of replications in a non-radioactive medium. Therefore, after 3 duplications, exactly two DNA molecules will contain radioactive thymidine.

162) Theory based

163) **Explaining the Question:**
Asking about sequence of m-RNA that produced by given DNA.

Concept:
Transcription

Solution:
Transcription occurs from the promotor towards terminator. In the diagram, this direction is from right to left. RNA polymerase transcribed 3 → 5 strand of DNA.
3 → 5 strand of DNA → 3' - TCGTA - 5'
m-RNA synthesized 5' → 3' 5'-AGCAU - 3'.

Final Answer:
Option (2) 5'-AGCAU-3'

165) NCERT Pg. # 122

166)

Explain Question : The question asks when DNA replication occurs in E. coli, a prokaryotic organism.
Concept : DNA replication.
Explanation : In prokaryotes like E. coli, DNA replication occurs continuously and is not restricted to a particular phase, unlike eukaryotic cells.
Final answer : No such specific phase

167) **Explain Question :** The question asks about the historical belief regarding the nature of genetic material before DNA was recognized as the hereditary material.

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Concept : Search Of Genetic Material

Solution: Prior to the work of Oswald Avery, Colin Macleod, and Maclyn McCarty (1933-44), the genetic material was widely thought to be protein. This was because proteins were believed to be more complex and diverse than nucleic acids, which led many scientists to favor proteins as the carriers of genetic information.

Final answer : Protein

170) NCERT-XII Pg. No. 106

172) **Solution:**

A. **Degeneracy**: multiple codons code for same amino acid.
B. Variation usually occurs at **third base (wobble position)**.

Final Answer: (2)

173) **Solution:** The original spliced mRNA codes for 150 amino acids. An unspliced intron adds 90 nucleotides, which codes for 30 additional amino acids (90 bases / 3 = 30). The total length becomes 180 amino acids. Eukaryotic nuclear-encoded mRNA is translated by 80S ribosomes located in the cytoplasm or on the rough ER.
Final Answer : (1)

174) Ans: 2003

175) Ans: 216

176) **Correct Answer: Option (1)**
Explanation: High-throughput technologies (RNA-seq, microarrays) enable genome-wide or transcriptome-wide parallel analysis of thousands of genes simultaneously, moving beyond the single-gene scale of conventional methods.

177) **Correct Answer: Option (2) — Three**
Explanation: Statements (i), (ii), and (vi) are correct. A polyribosome (polysome) is formed when multiple ribosomes attach to a single mRNA to simultaneously produce multiple copies of the identical polypeptide. Polysomes are found in both prokaryotes and eukaryotes.

178) The operon will not be functional at all.

179) Correct Answer: Option (2) — (i), (ii) and (iii) only
Detailed Explanation:
Statements (i), (ii), and (iii) are correct: Direct verbatim facts from NCERT Class 12 Biology. Sanger developed the sequencing principle, reads were aligned using overlapping fragments, and alignment required specialized computer algorithms.
Statement (iv) is incorrect: Frederick Sanger developed the dideoxy chain-termination method

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for DNA sequencing (and amino acid sequencing of proteins), not the bioinformatics computer programs for sequence alignment.

180) **Correct Option: (C) Z**

• Structure (X): Precursor oligosaccharide chain (Glucose – Galactose – N-acetylglucosamine – Galactose).

• Structure (Y): Terminal addition of Galactose to the H-antigen core, representing Blood Group B antigen.

• Structure (Z): Terminal addition of N-acetylgalactosamine (GalNAc) to the H-antigen core, which defines Blood Group A antigen.

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